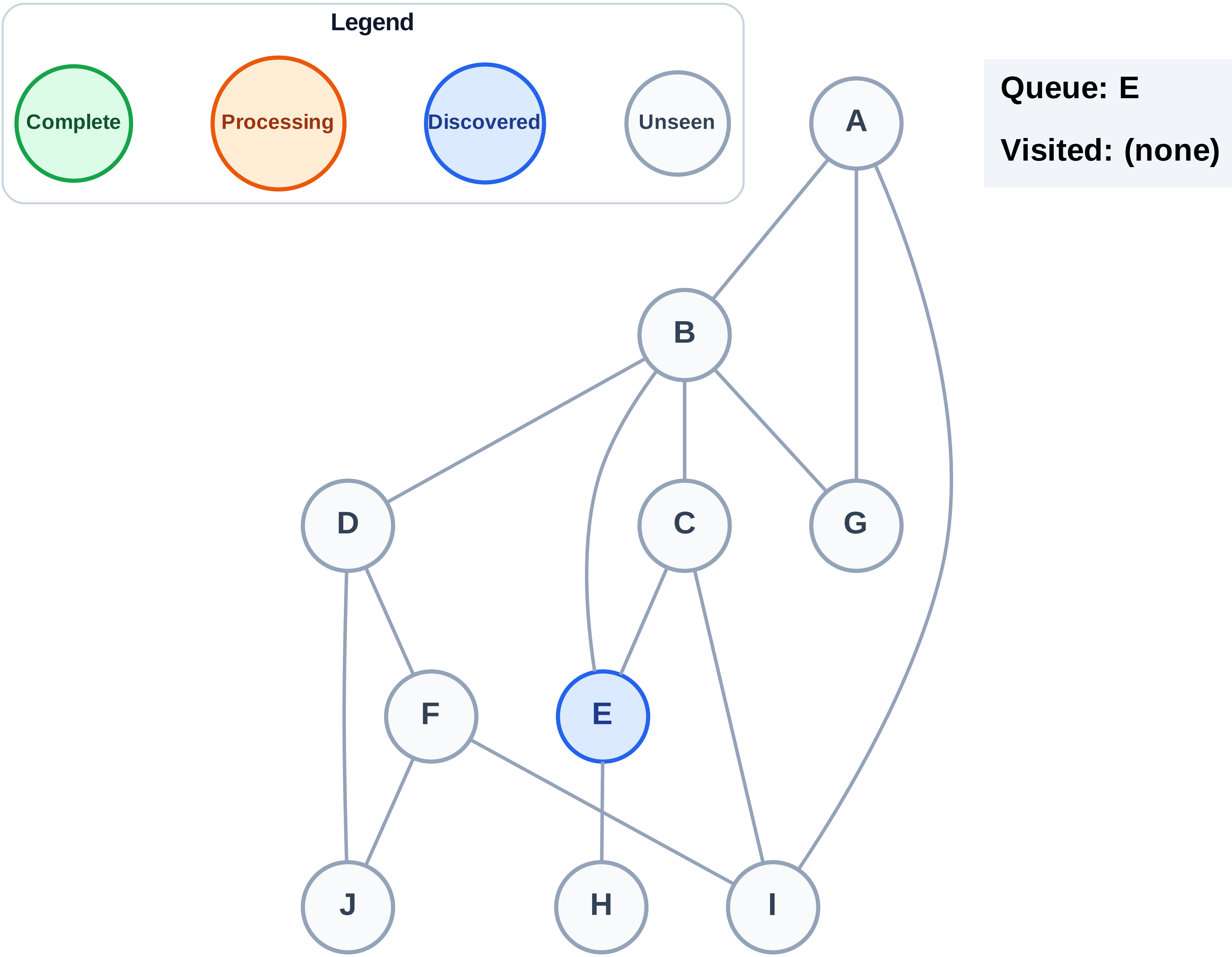
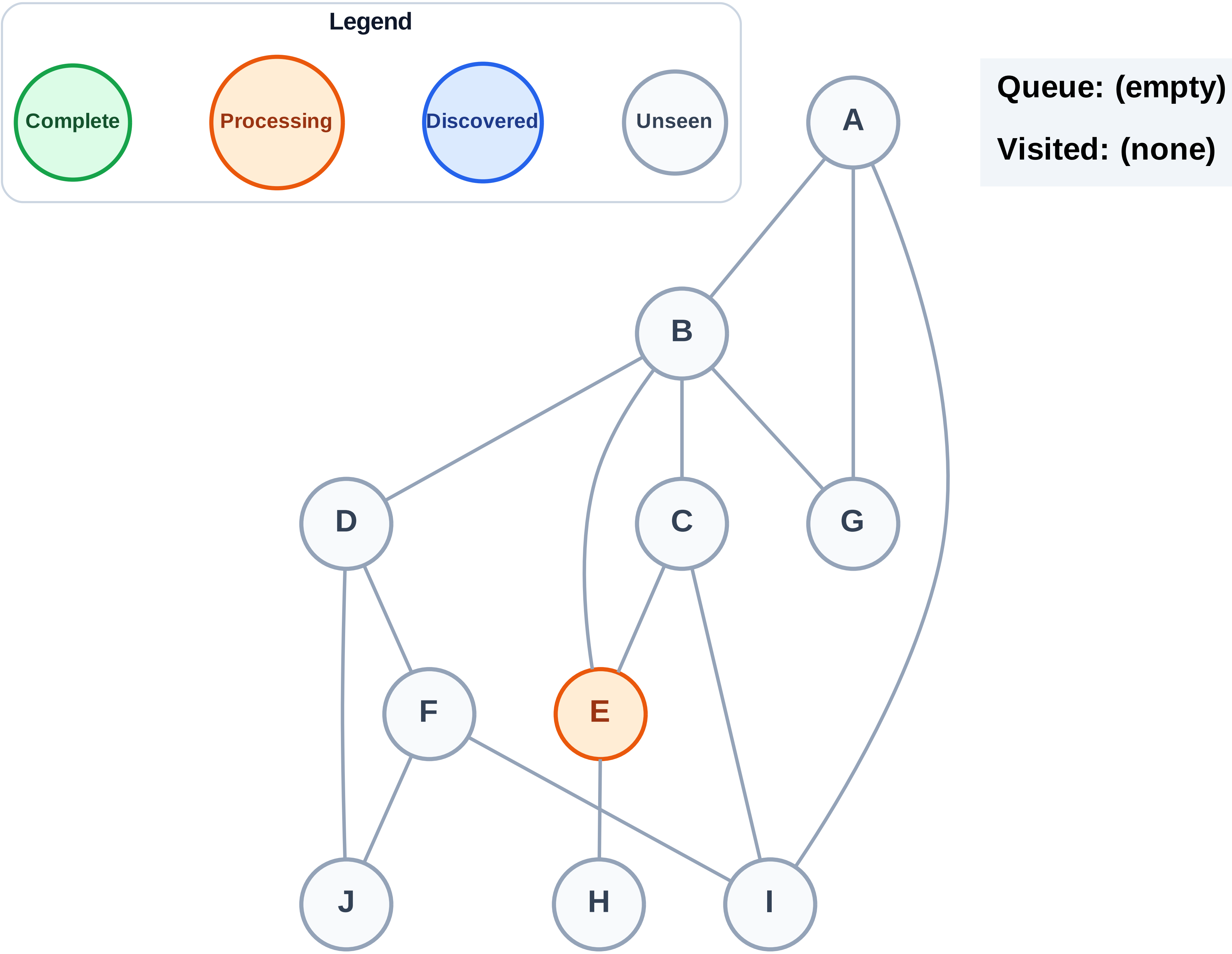
BFS Traversal

Step 1: Start at E



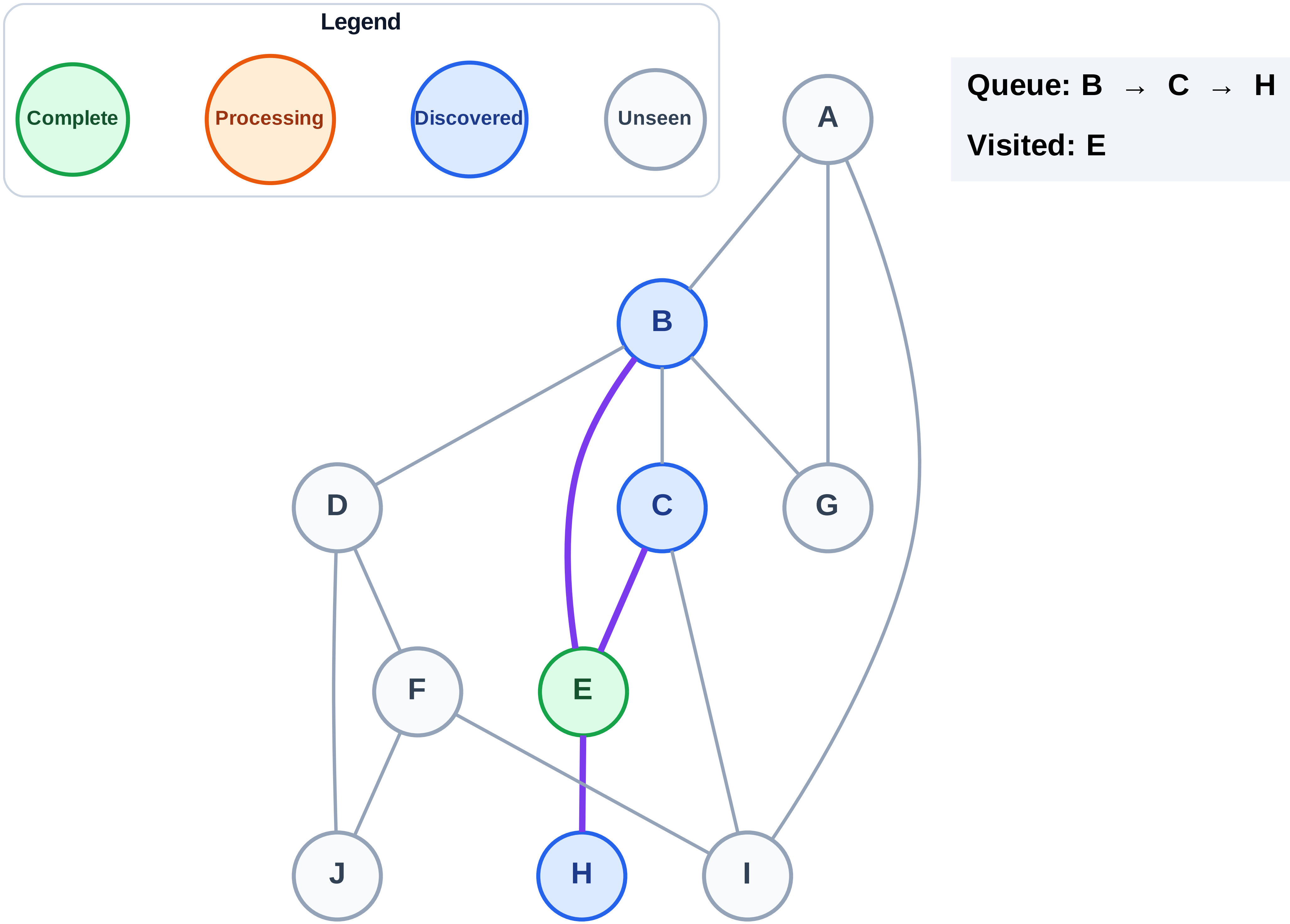
BFS Traversal

Step 2: Process node E



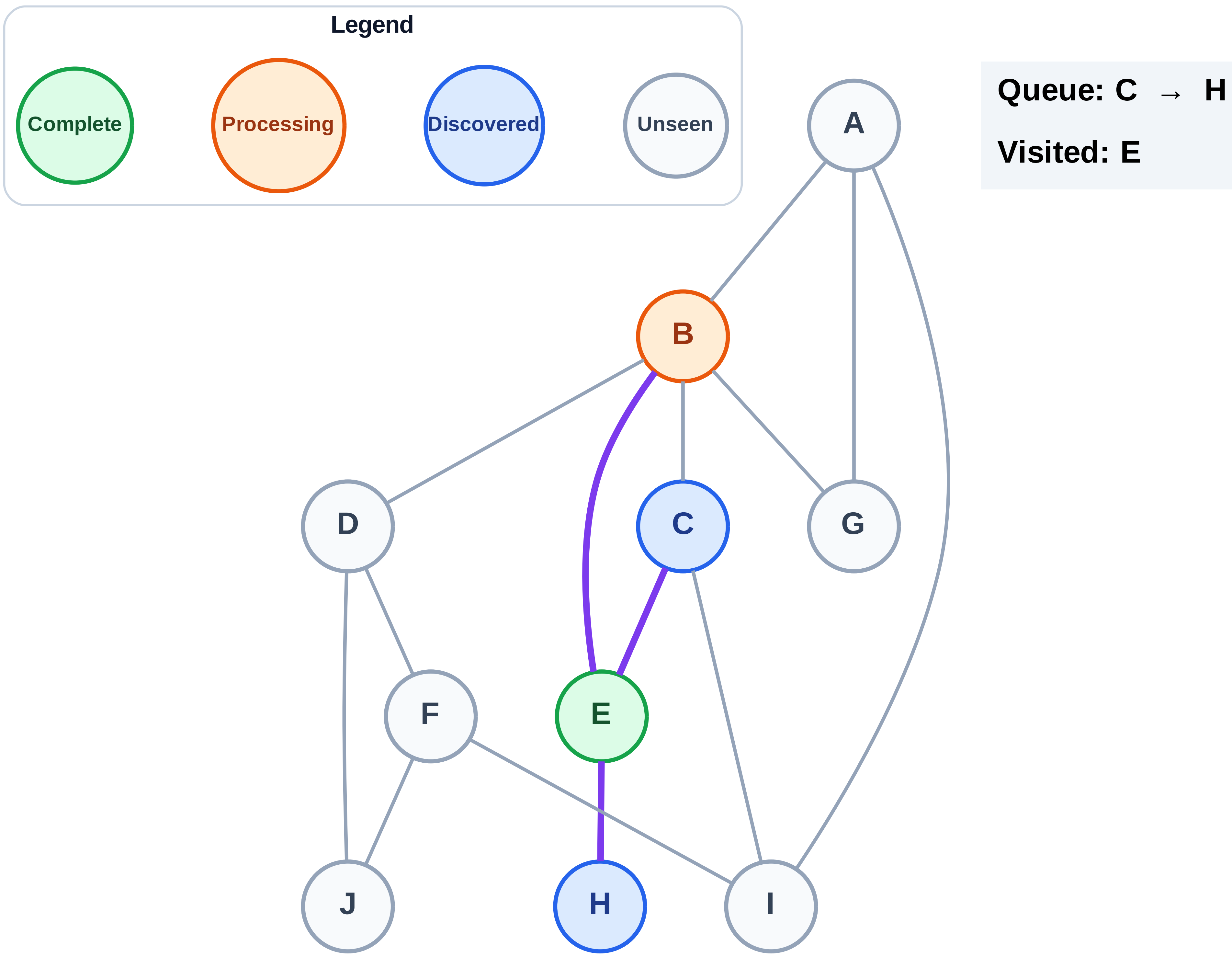
BFS Traversal

Step 3: Discover B, C, H from E



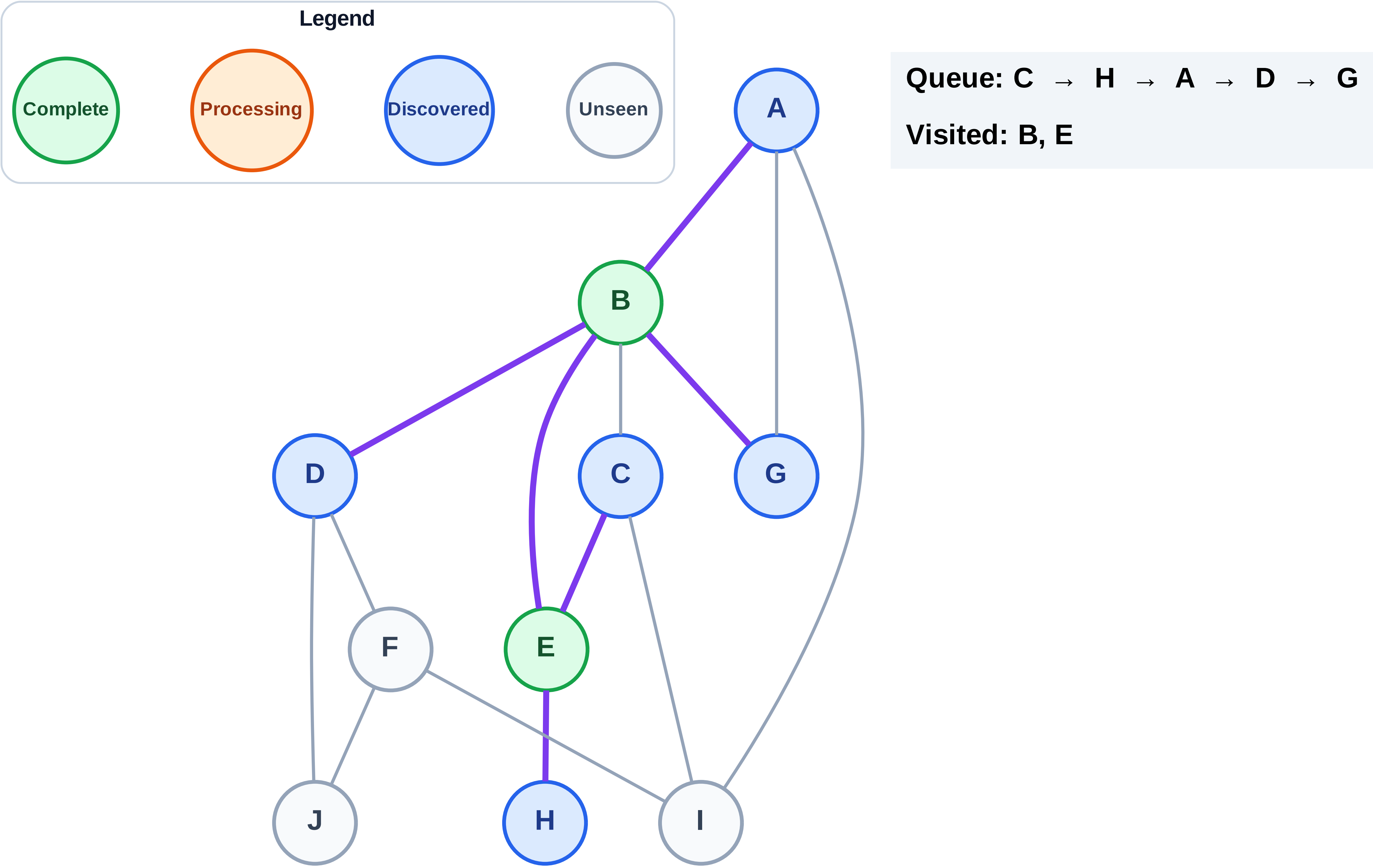
BFS Traversal

Step 4: Process node B



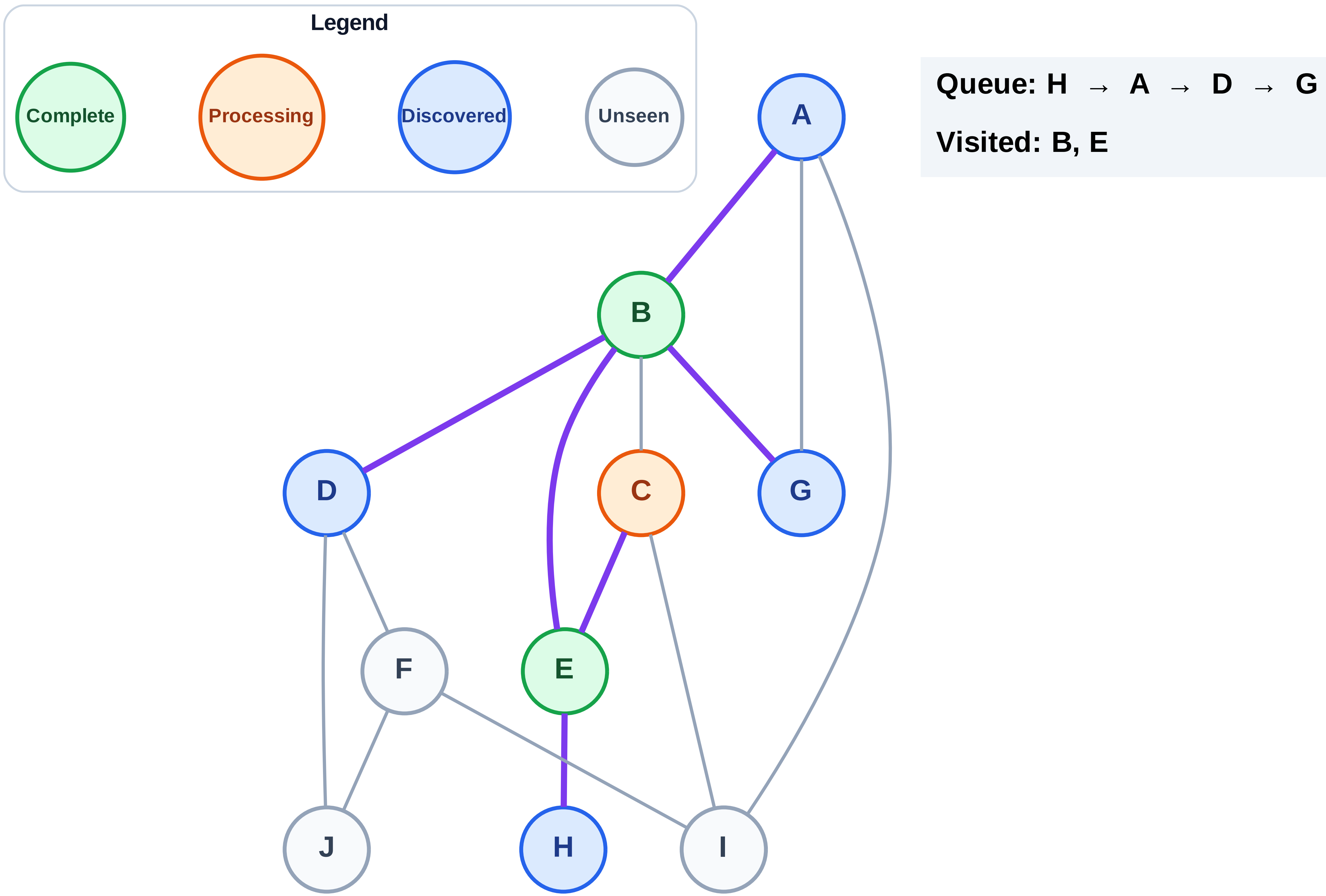
BFS Traversal

Step 5: Discover A, D, G from B



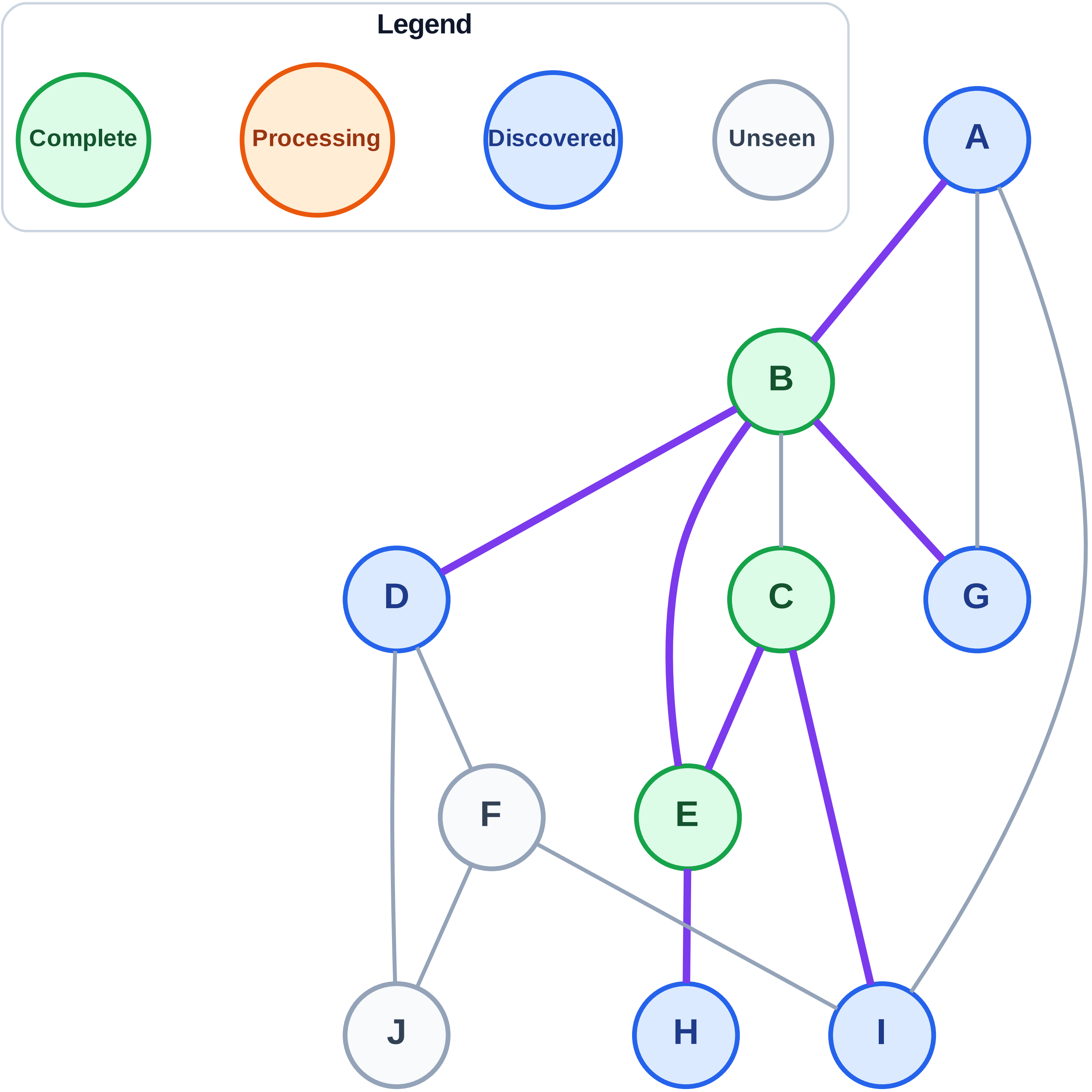
BFS Traversal

Step 6: Process node C



BFS Traversal

Step 7: Discover I from C

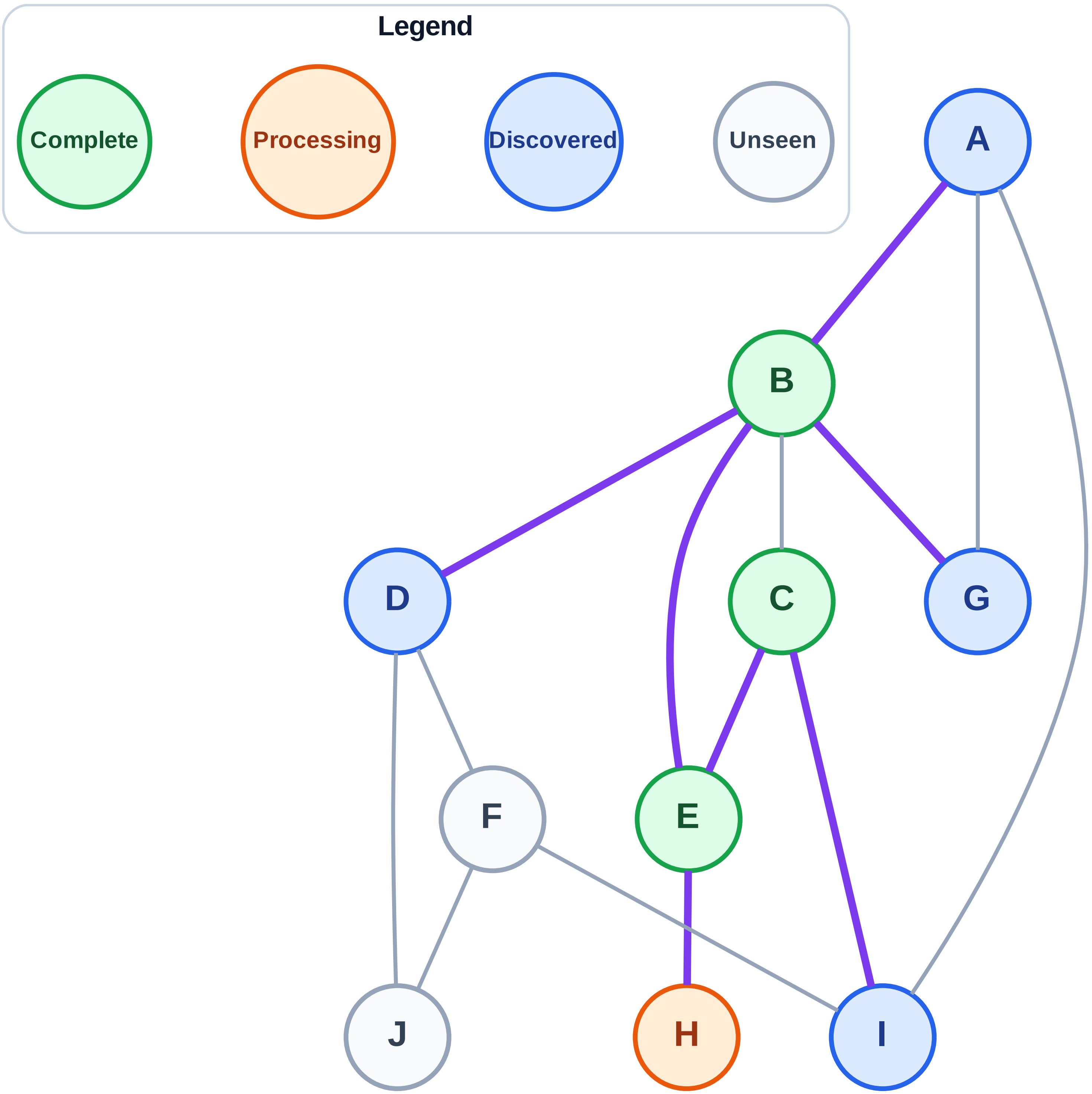


Queue: H → A → D → G → I

Visited: B, C, E

BFS Traversal

Step 8: Process node H

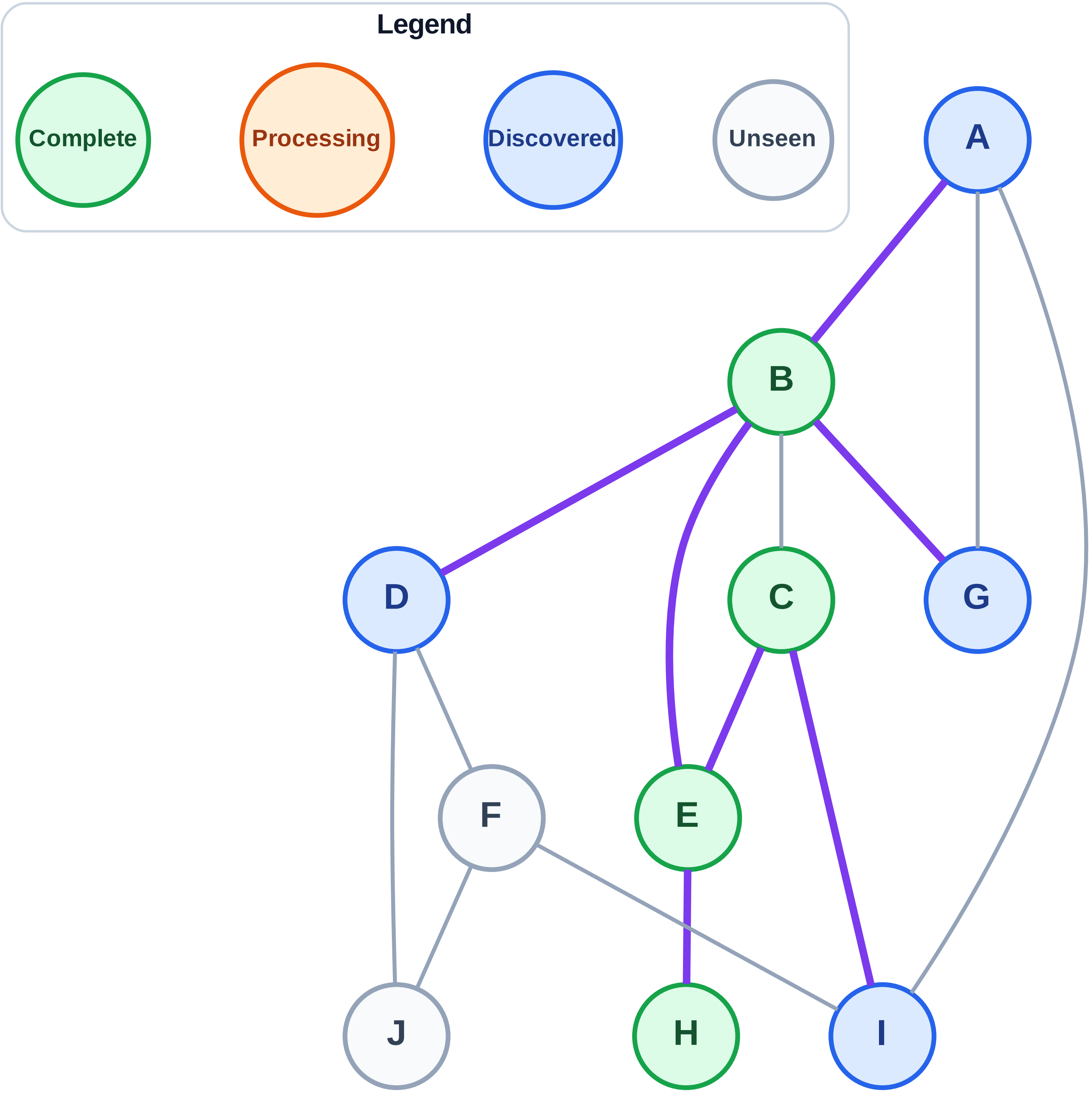


Queue: A → D → G → I

Visited: B, C, E

BFS Traversal

Step 9: No new neighbors from H

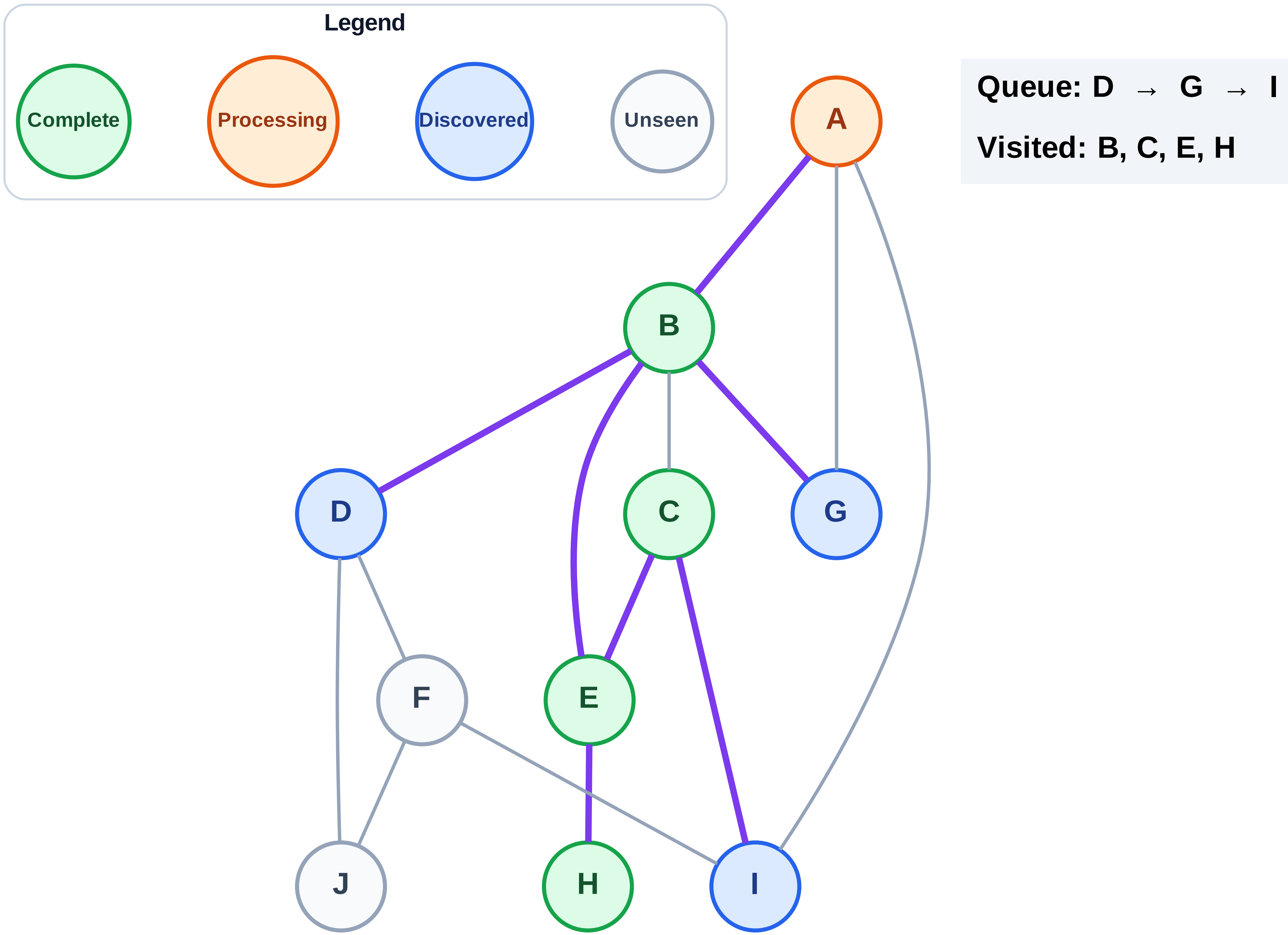


Queue: A → D → G → I

Visited: B, C, E, H

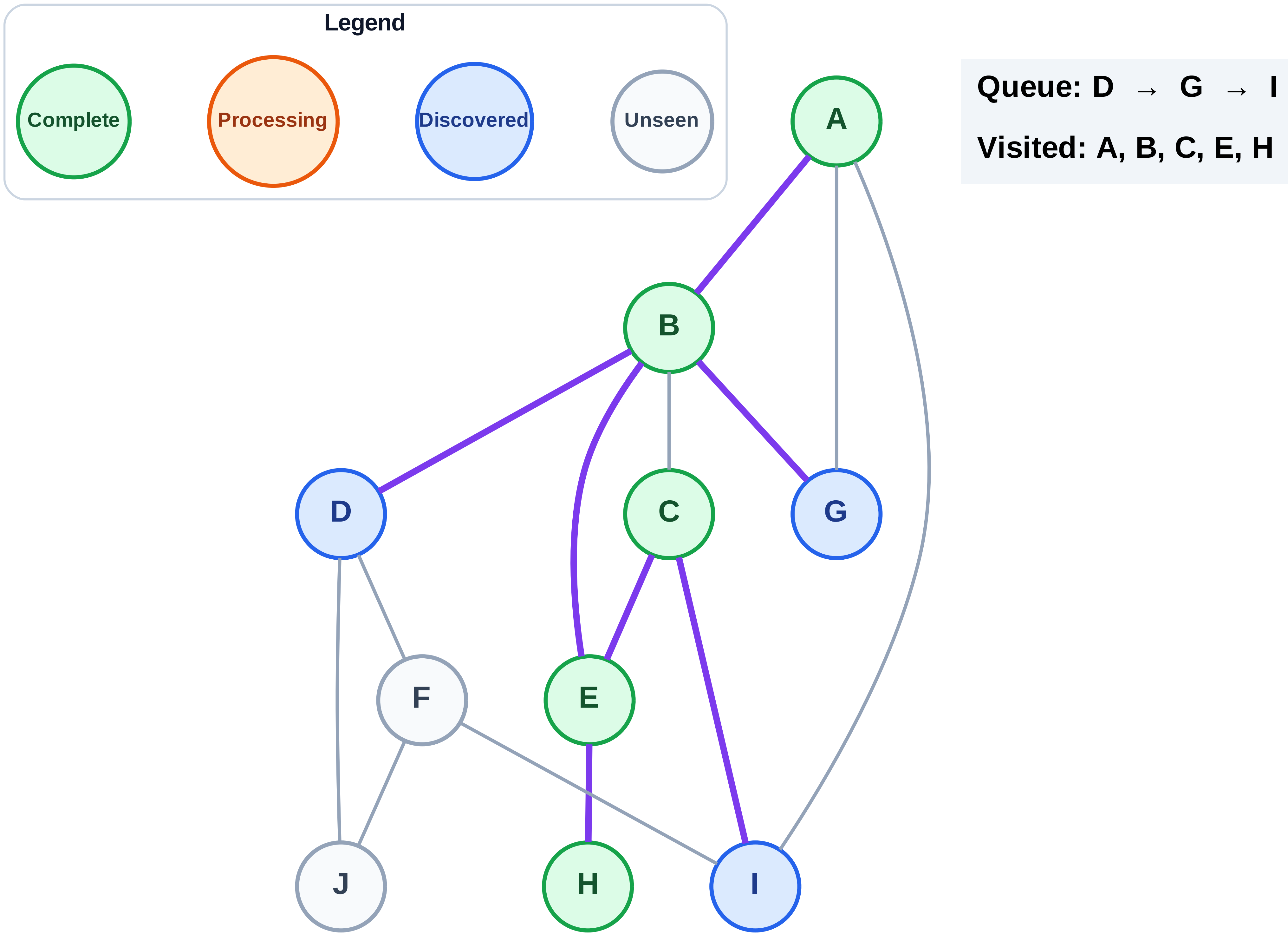
BFS Traversal

Step 10: Process node A



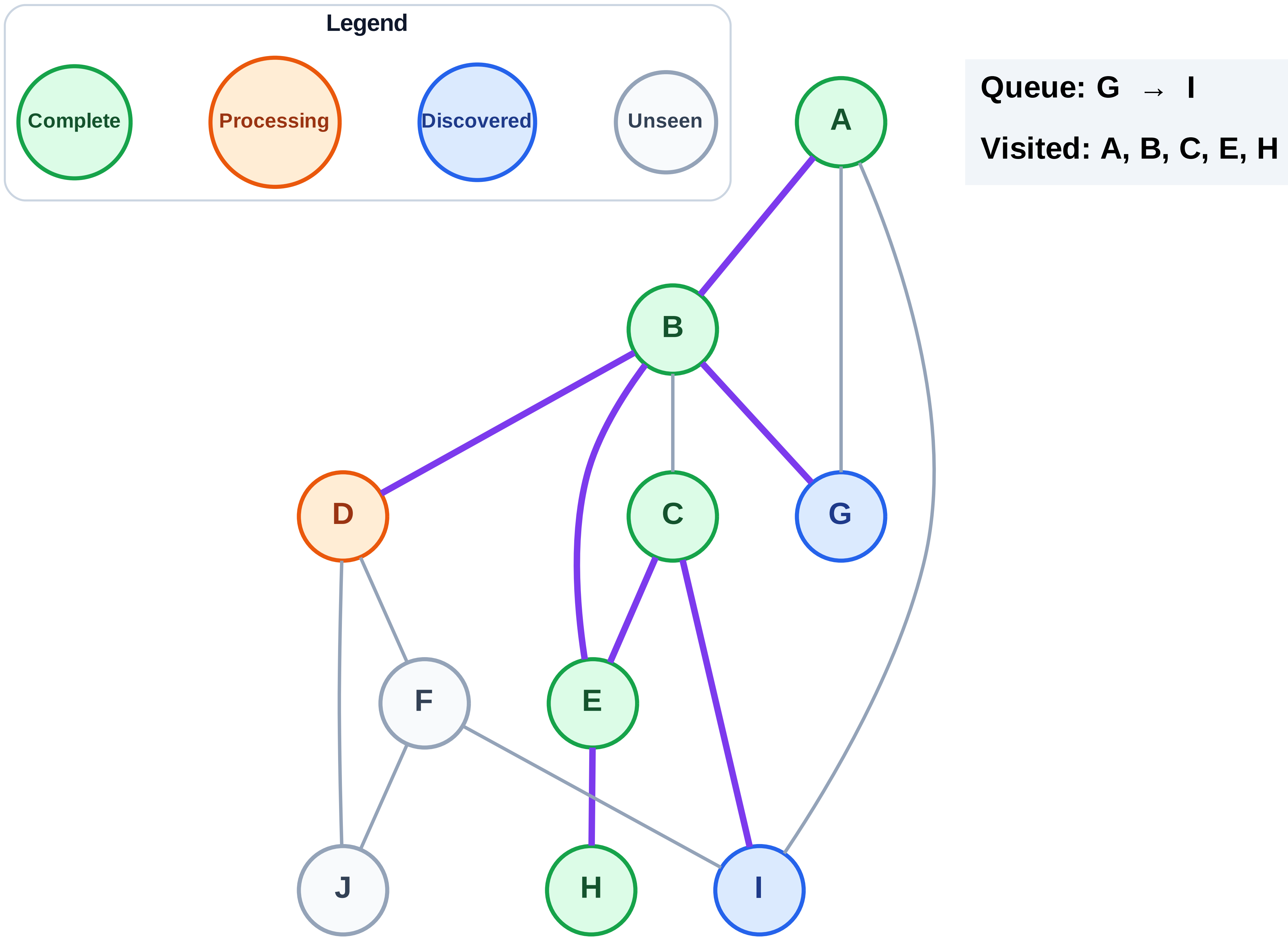
BFS Traversal

Step 11: No new neighbors from A



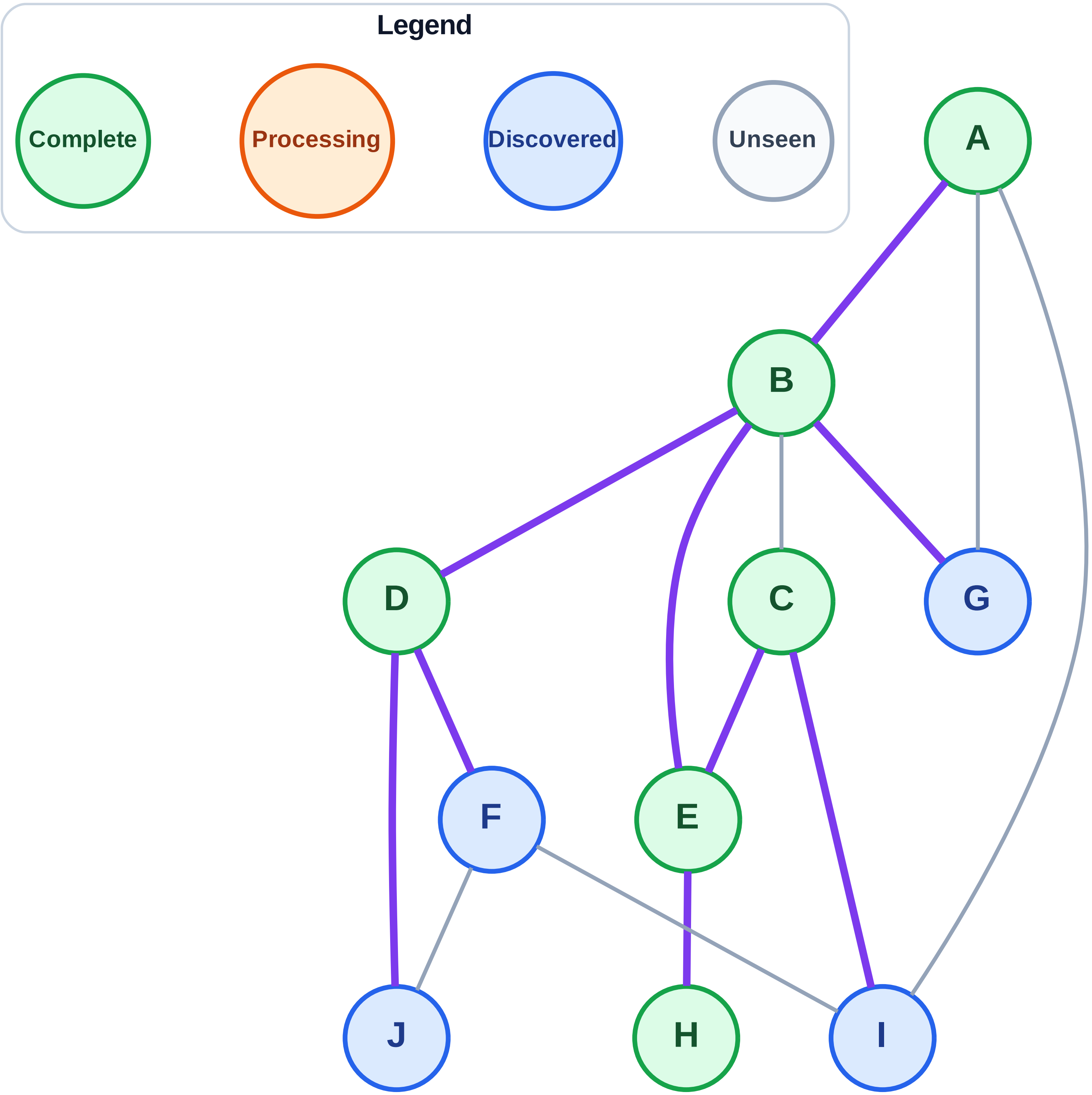
BFS Traversal

Step 12: Process node D



BFS Traversal

Step 13: Discover F, J from D

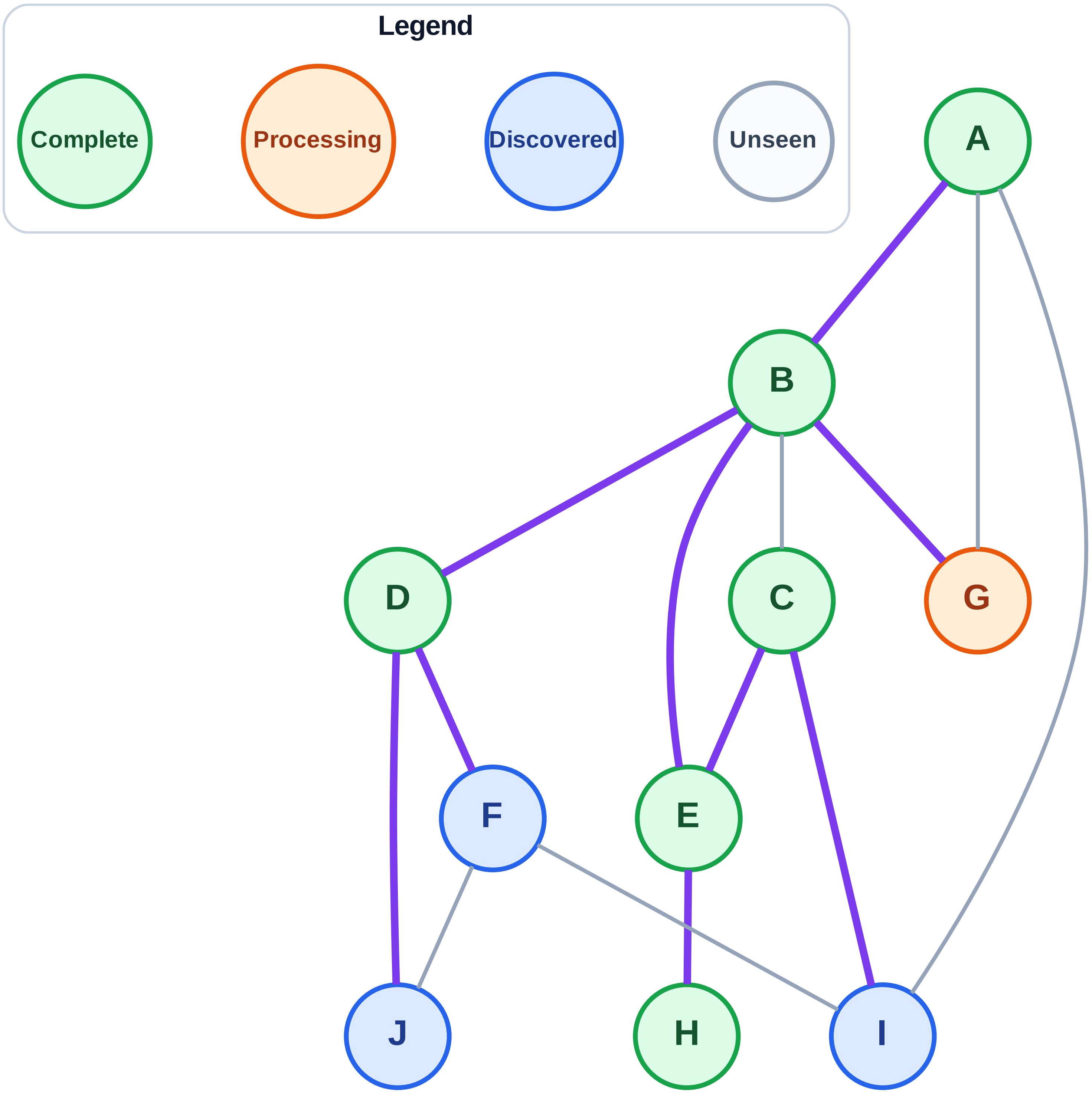


Queue: G → I → F → J

Visited: A, B, C, D, E, H

BFS Traversal

Step 14: Process node G

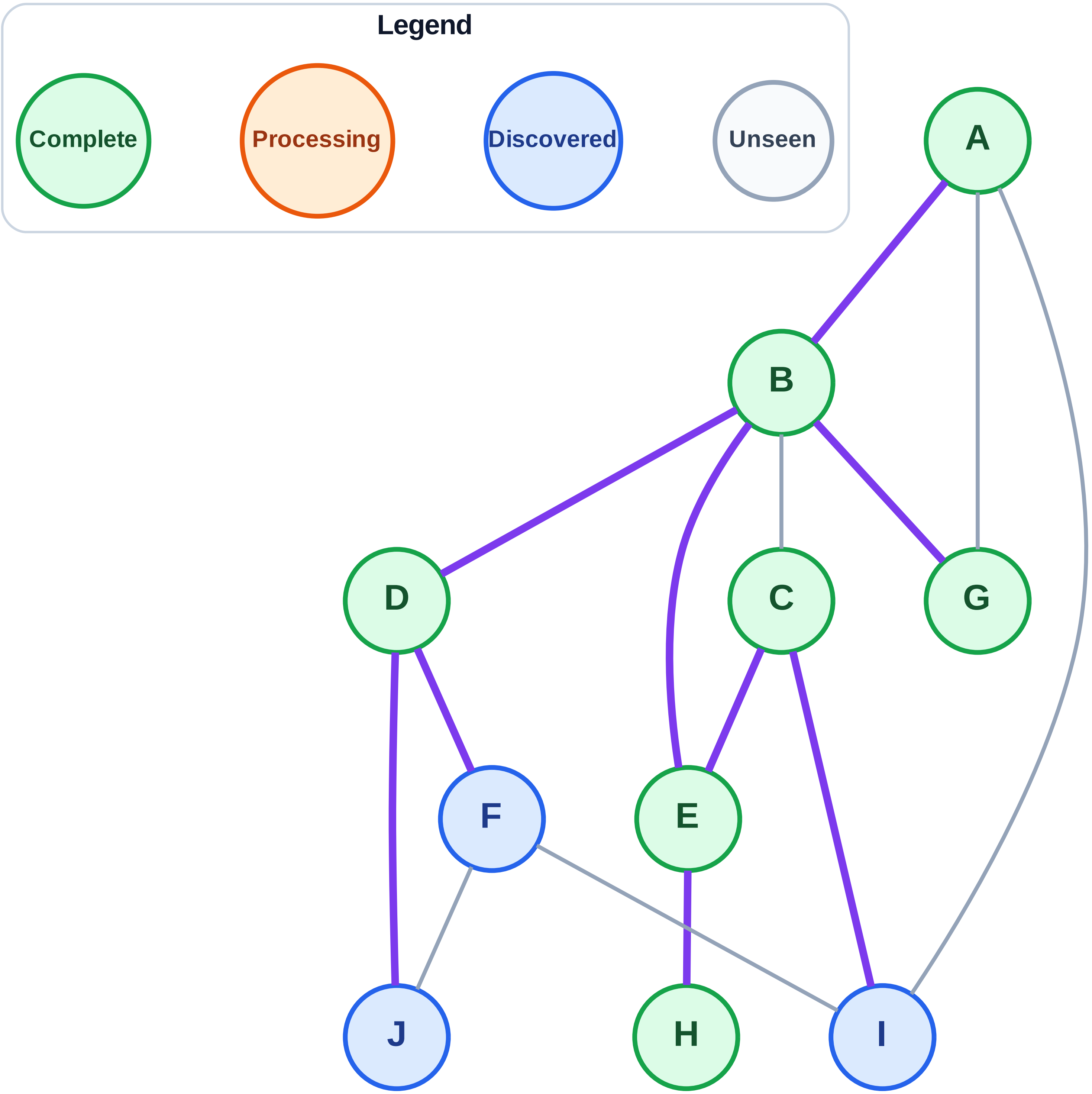


Queue: I → F → J

Visited: A, B, C, D, E, H

BFS Traversal

Step 15: No new neighbors from G

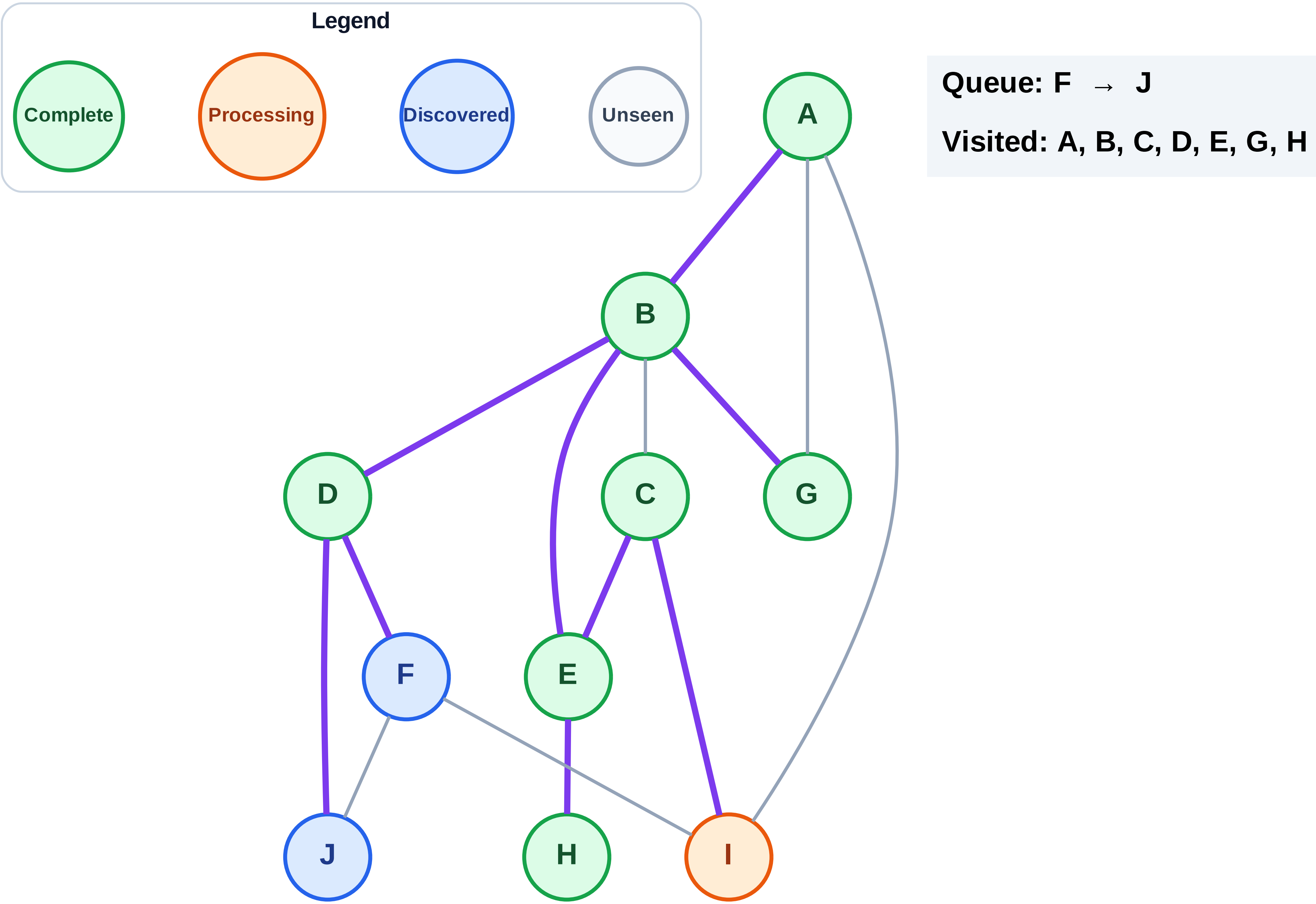


Queue: I → F → J

Visited: A, B, C, D, E, G, H

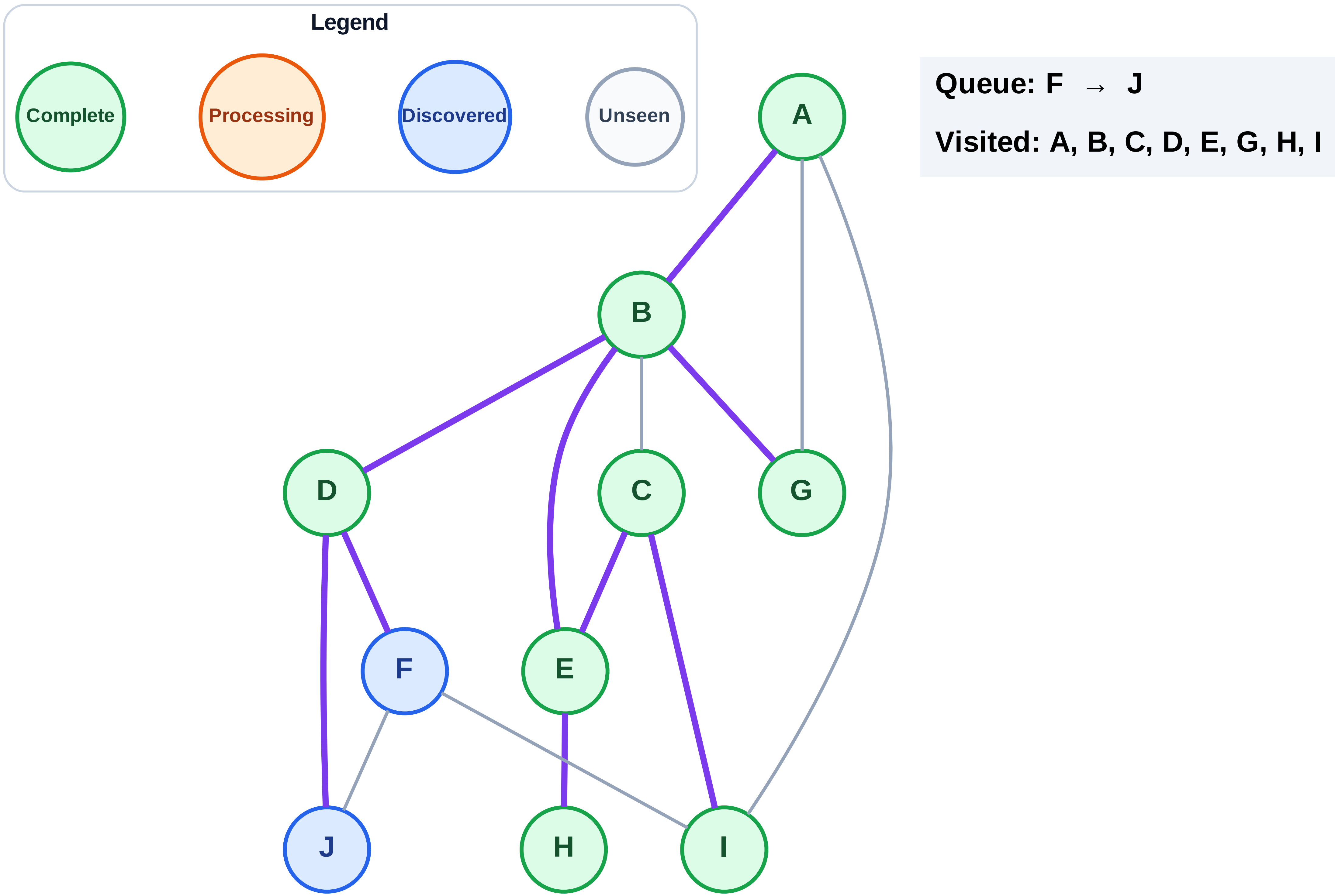
BFS Traversal

Step 16: Process node I



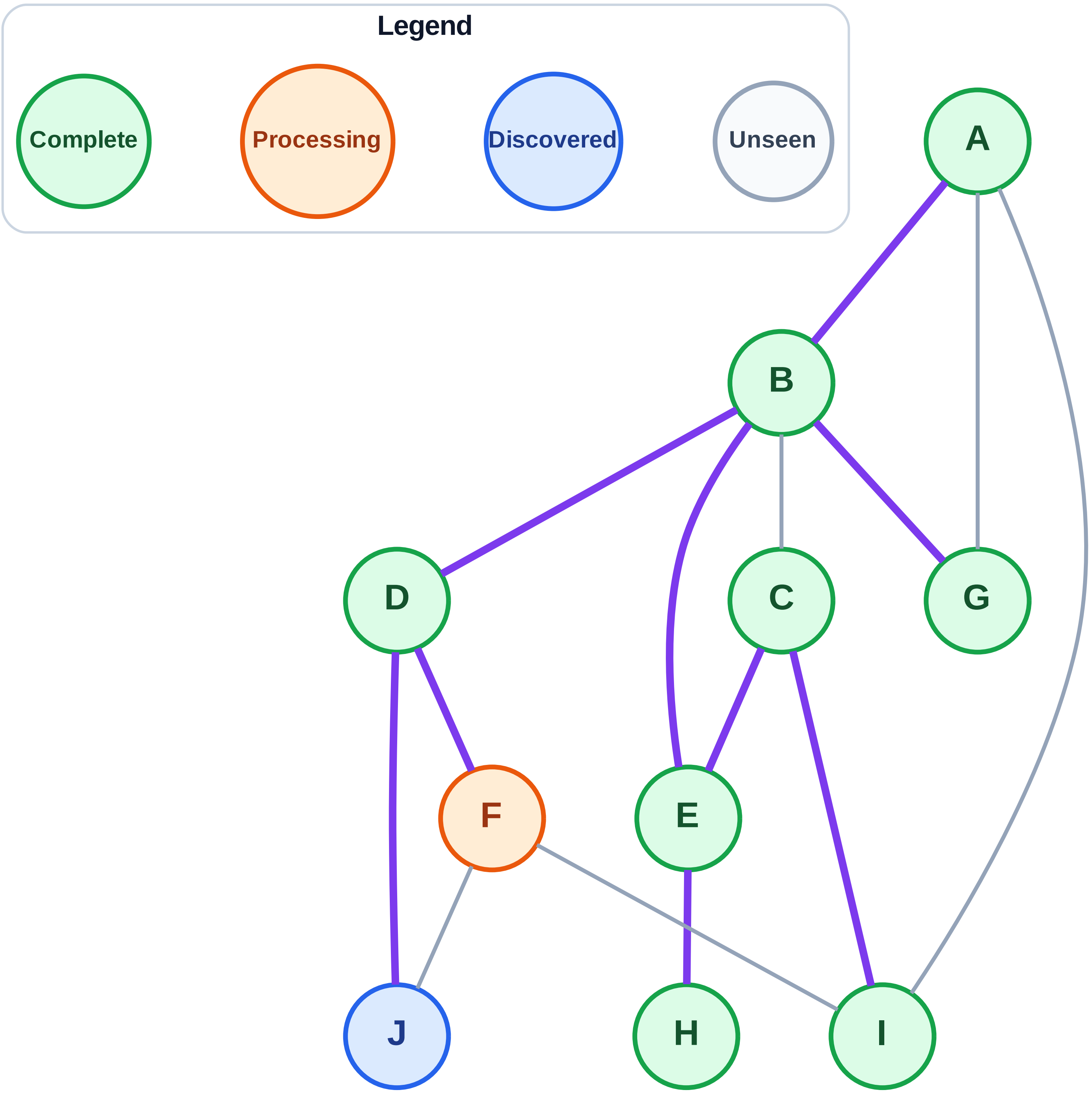
BFS Traversal

Step 17: No new neighbors from I



BFS Traversal

Step 18: Process node F

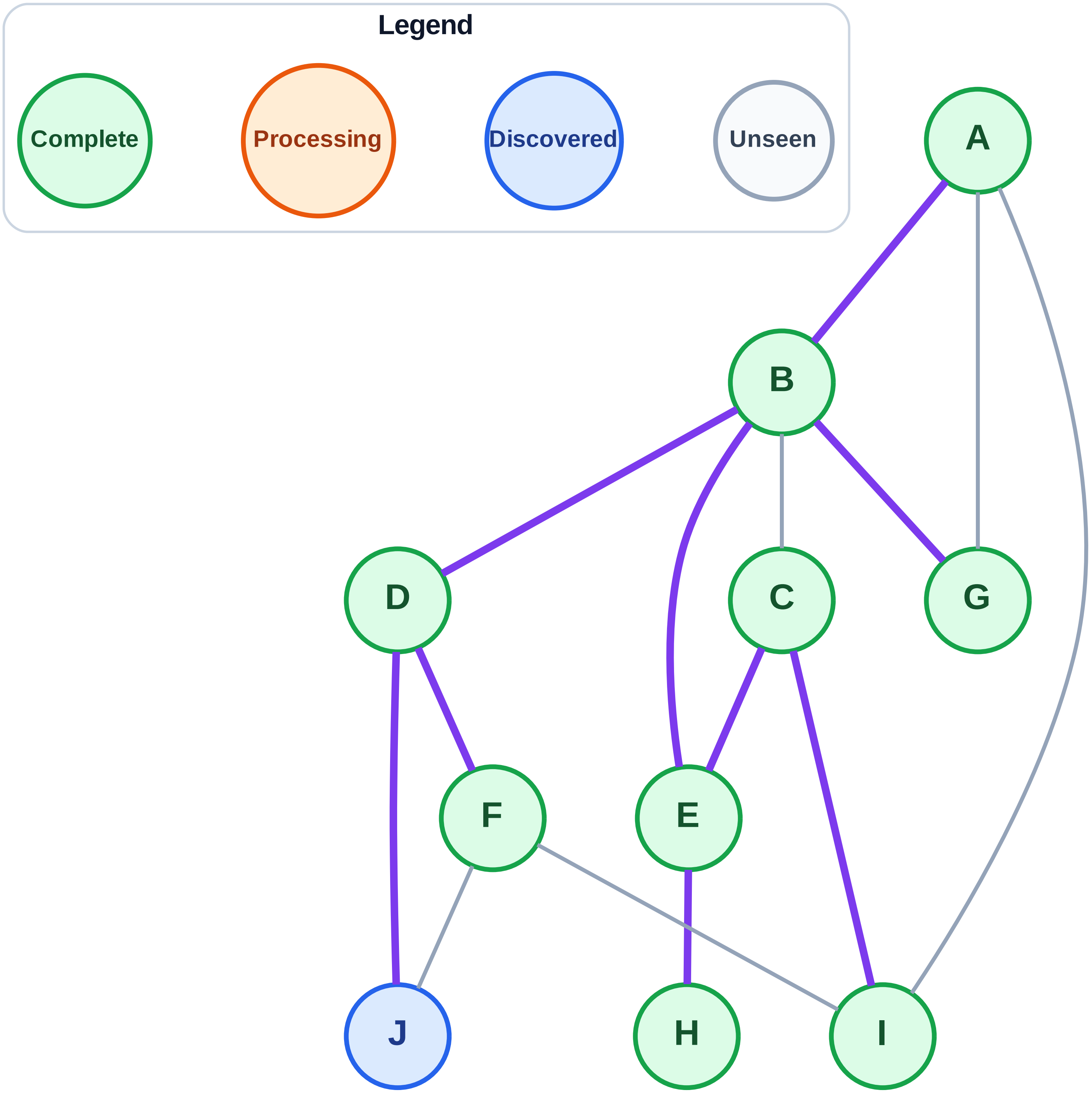


Queue: J

Visited: A, B, C, D, E, G, H, I

BFS Traversal

Step 19: No new neighbors from F

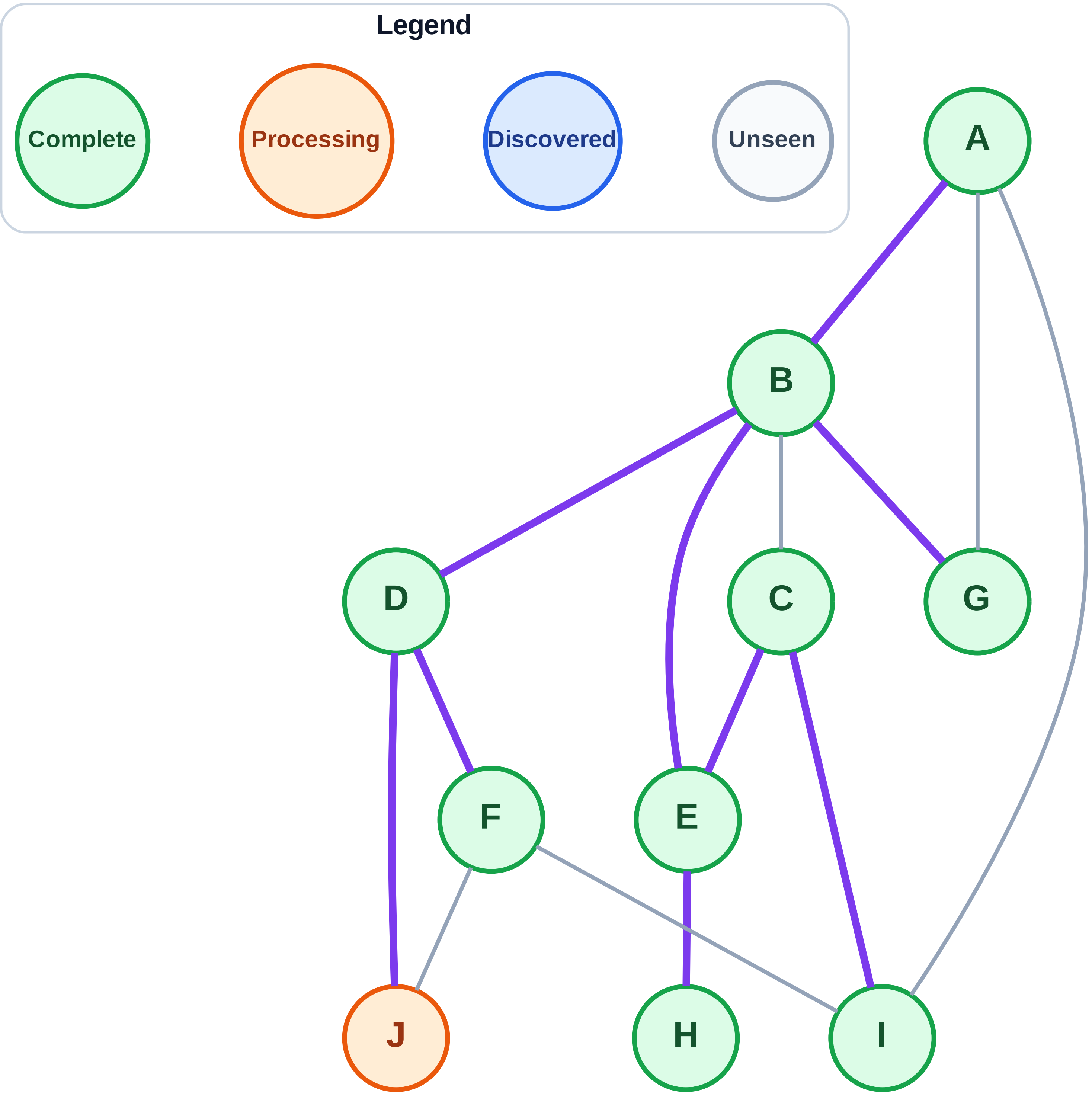


Queue: J

Visited: A, B, C, D, E, F, G, H, I

BFS Traversal

Step 20: Process node J

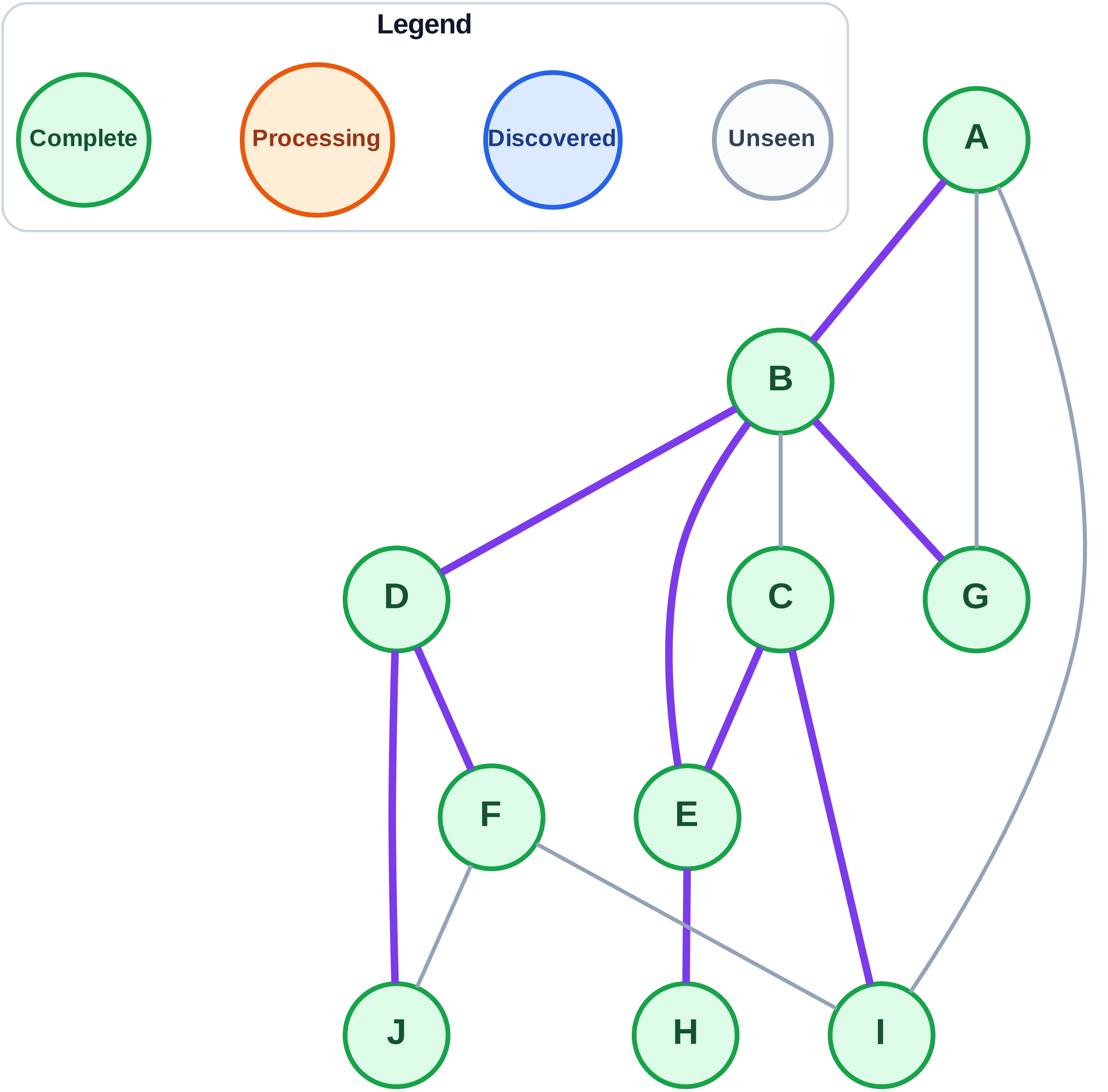


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I

BFS Traversal

Step 21: No new neighbors from J

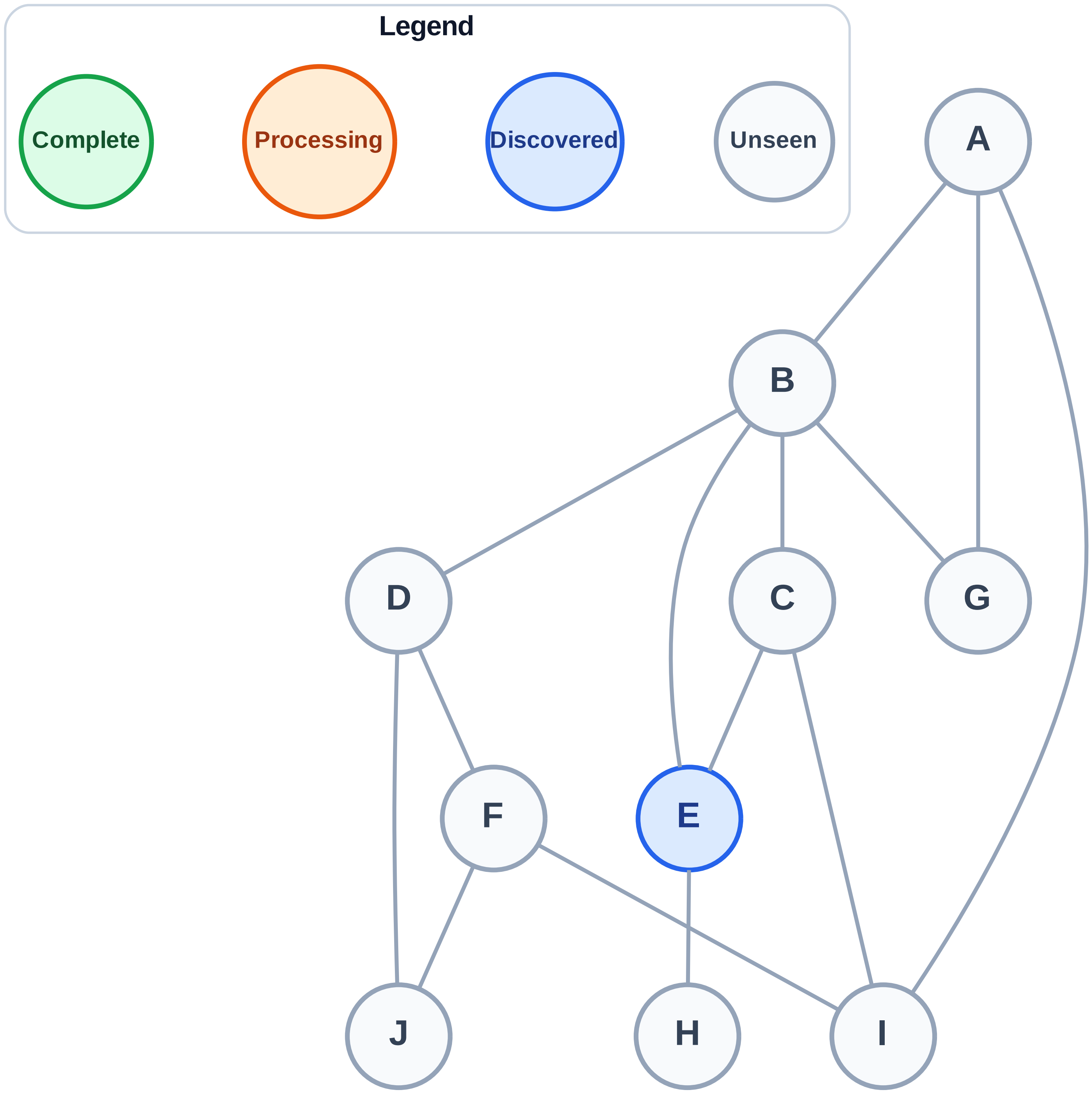


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

DFS Traversal

Step 1: Start at E

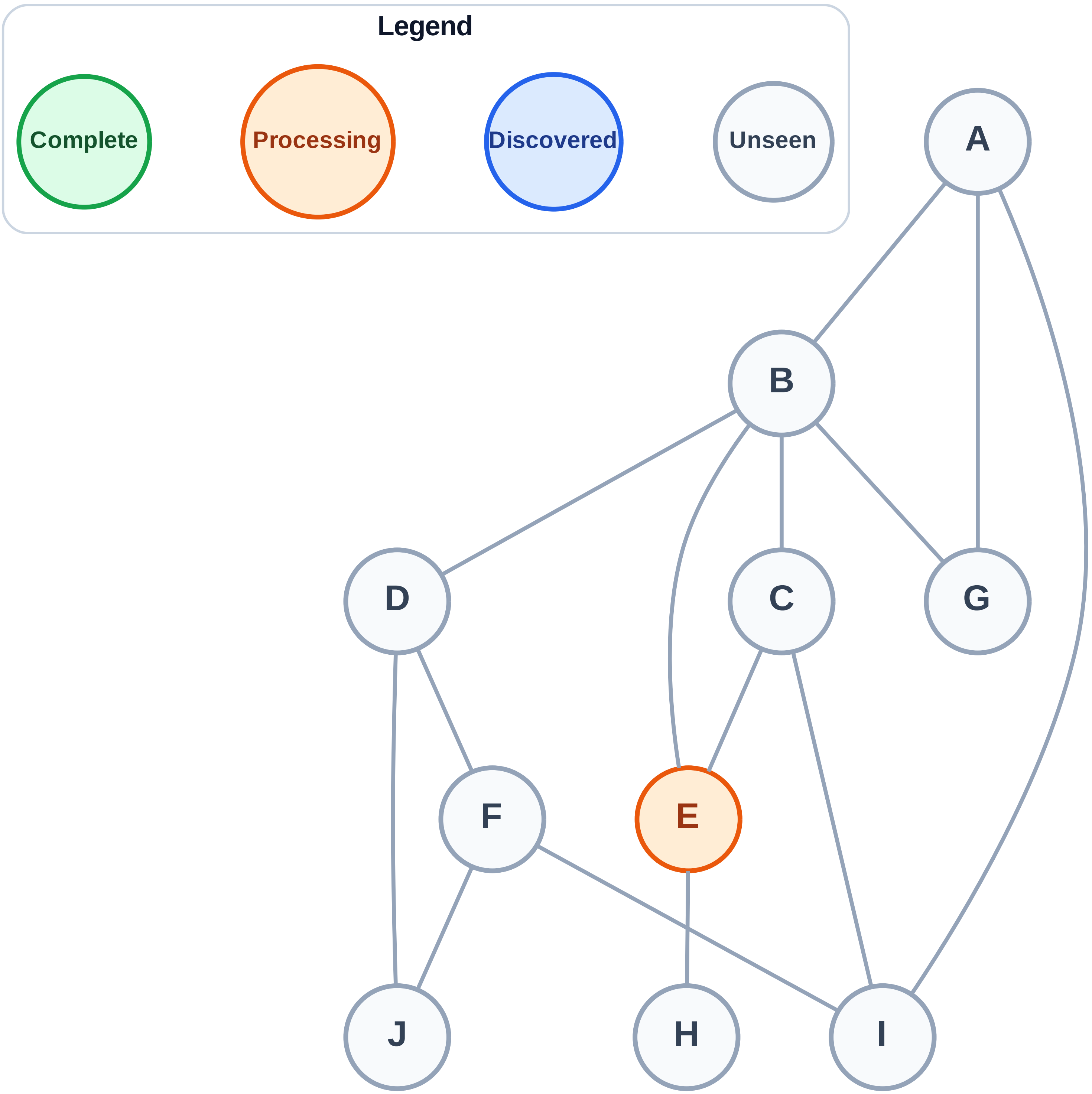


Stack (top at right): E

Visited: (none)

DFS Traversal

Step 2: Process node E

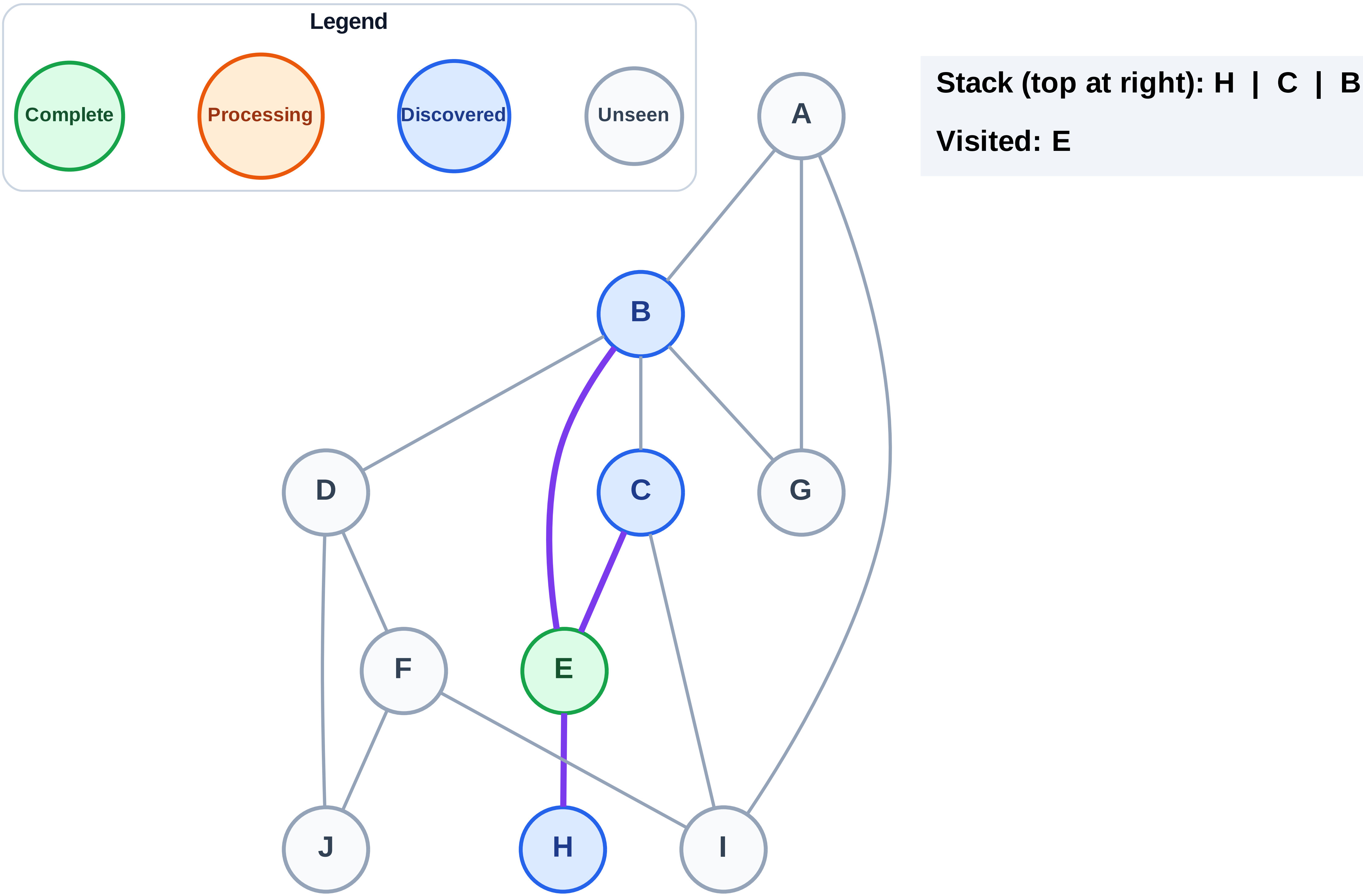


Stack (top at right): (empty)

Visited: (none)

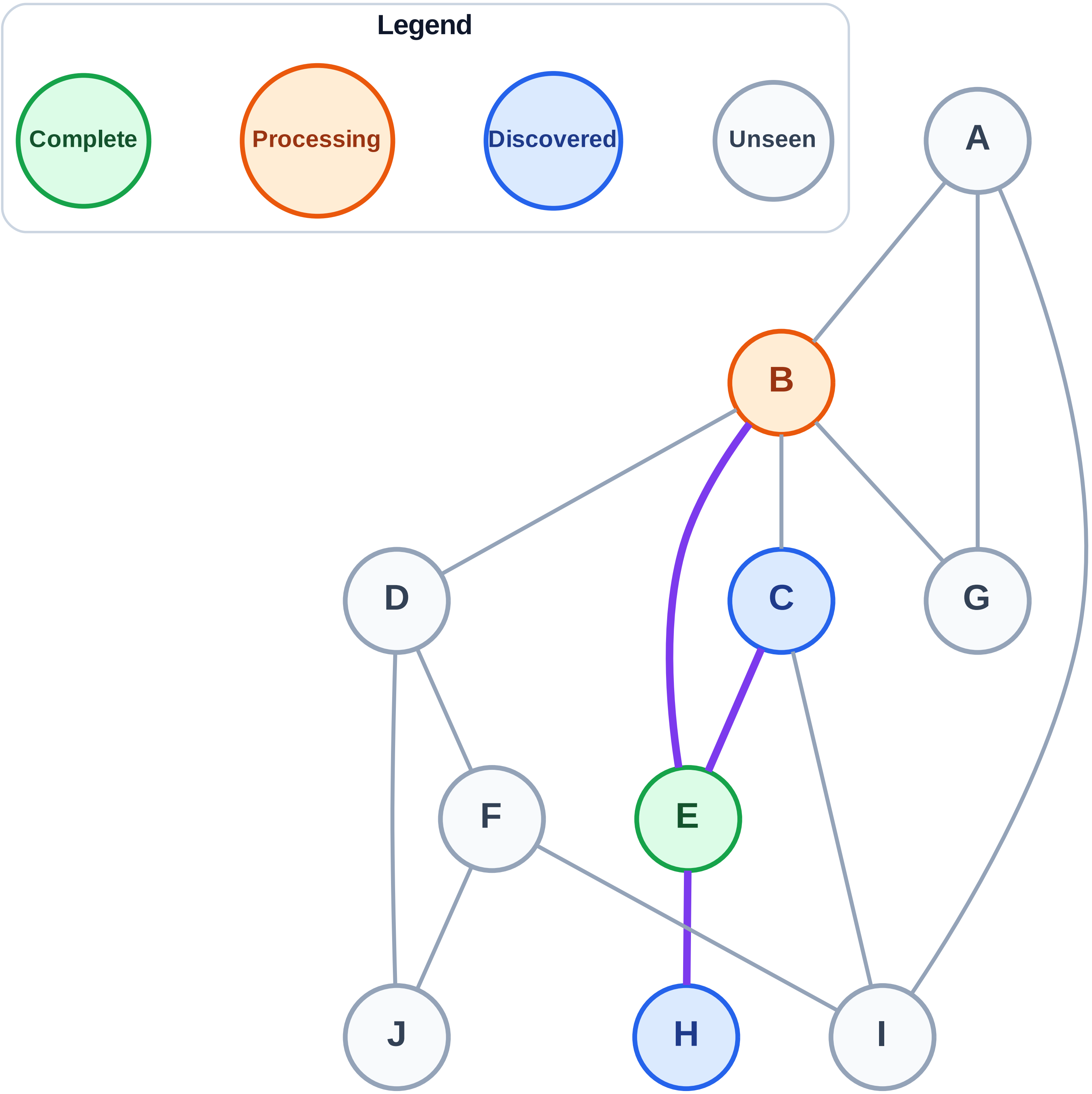
DFS Traversal

Step 3: Discover H, C, B from E



DFS Traversal

Step 4: Process node B

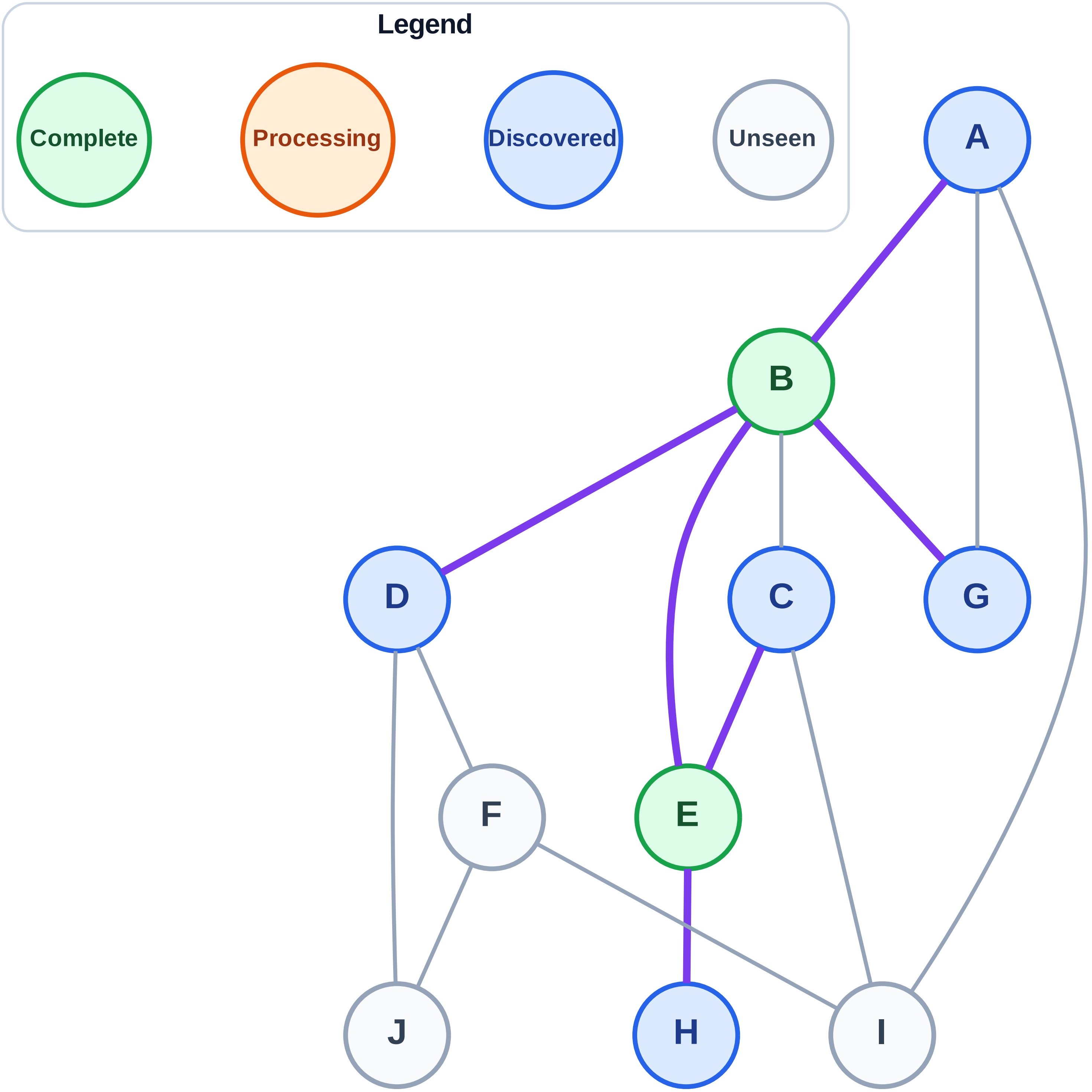


Stack (top at right): H | C

Visited: E

DFS Traversal

Step 5: Discover G, D, A from B

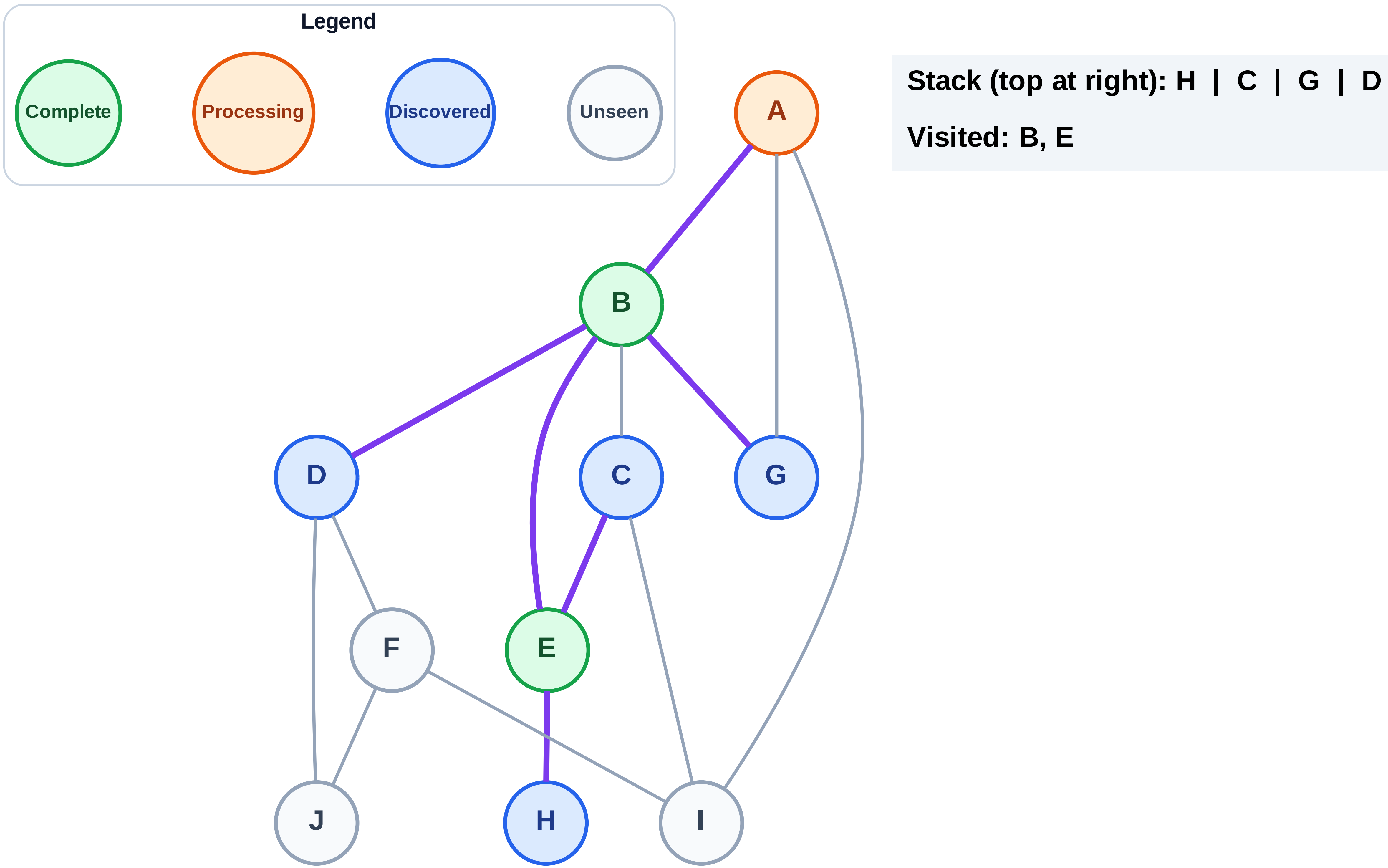


Stack (top at right): H | C | G | D | A

Visited: B, E

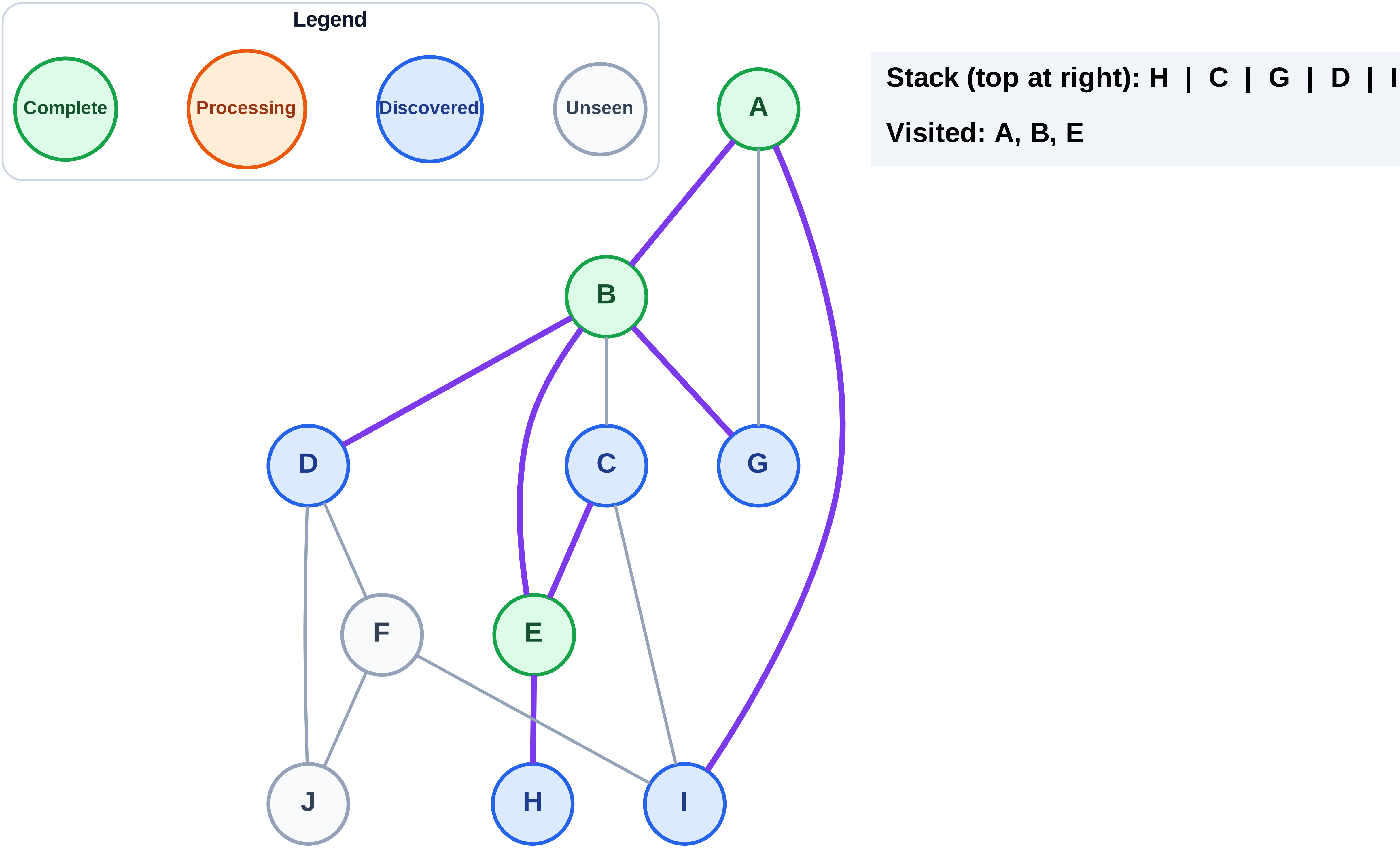
DFS Traversal

Step 6: Process node A



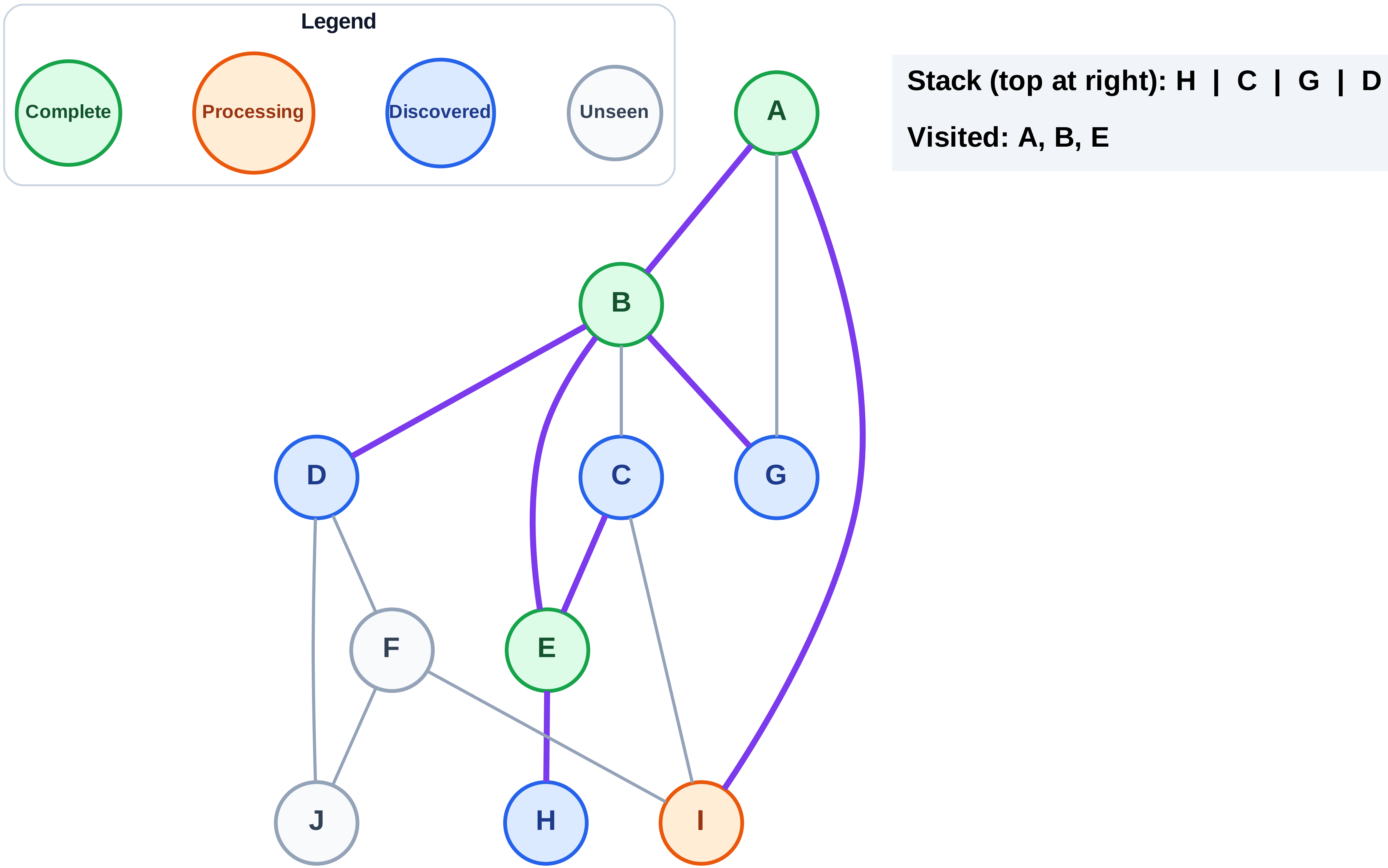
DFS Traversal

Step 7: Discover I from A



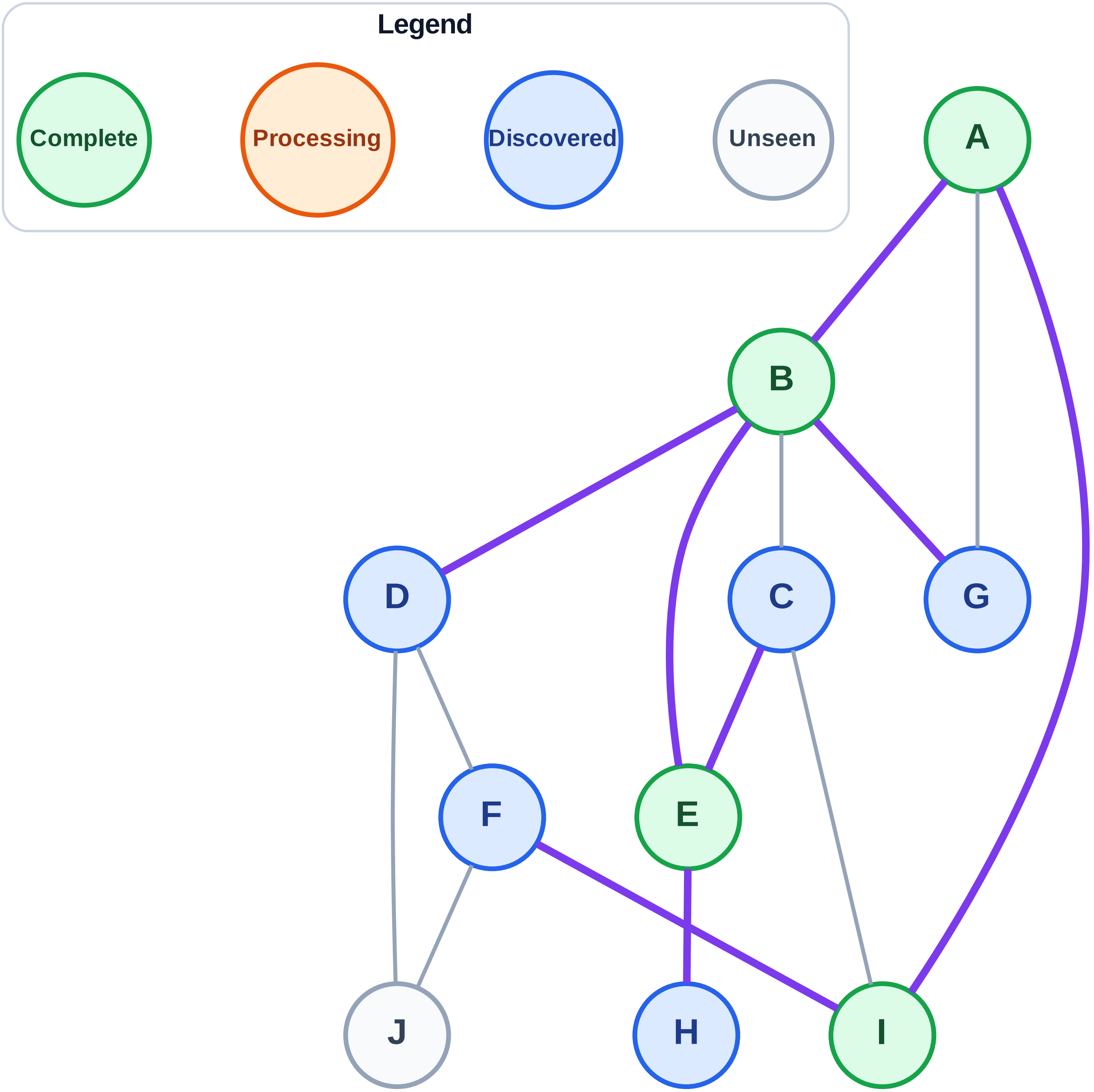
DFS Traversal

Step 8: Process node I



DFS Traversal

Step 9: Discover F from I

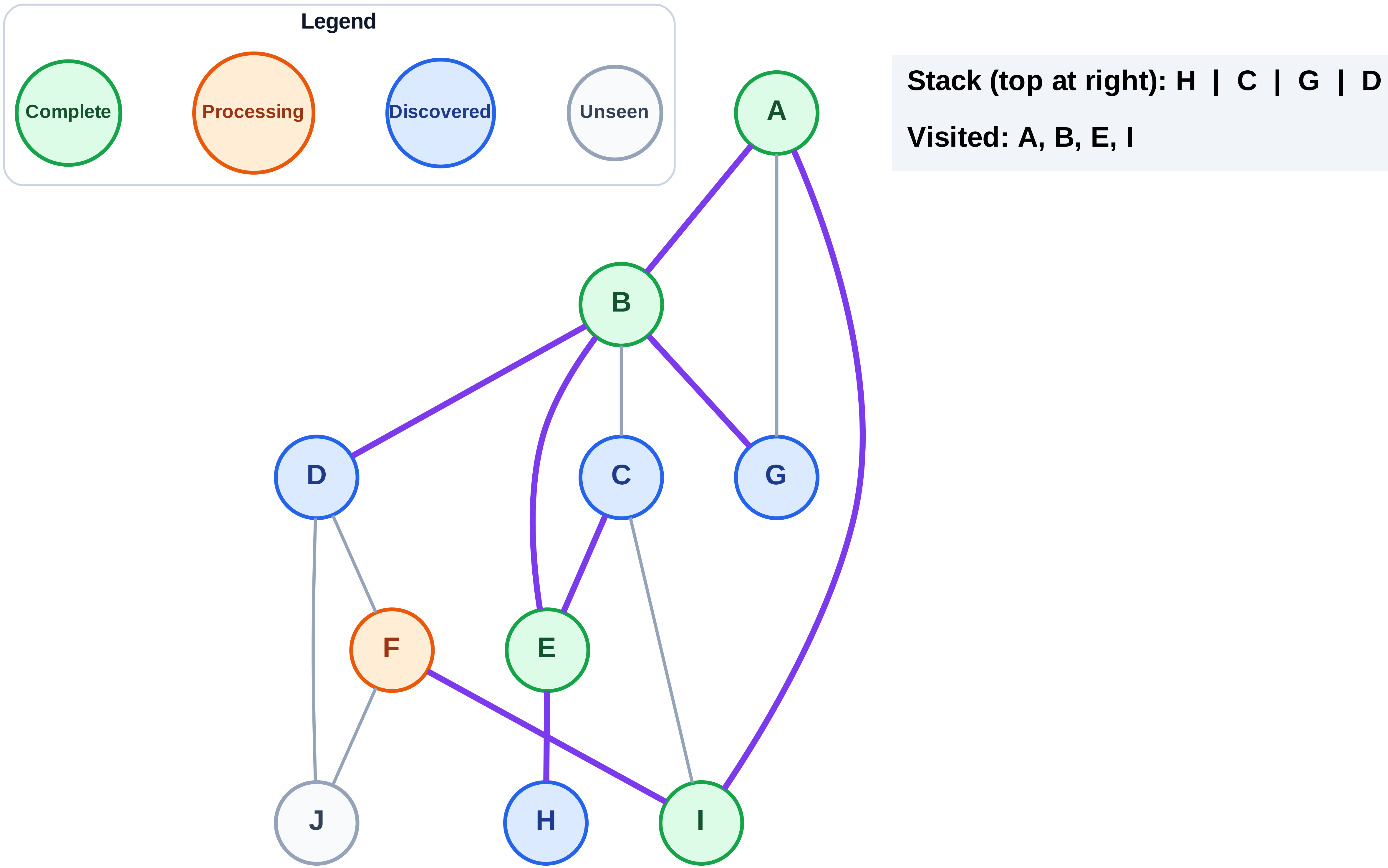


Stack (top at right): H | C | G | D | F

Visited: A, B, E, I

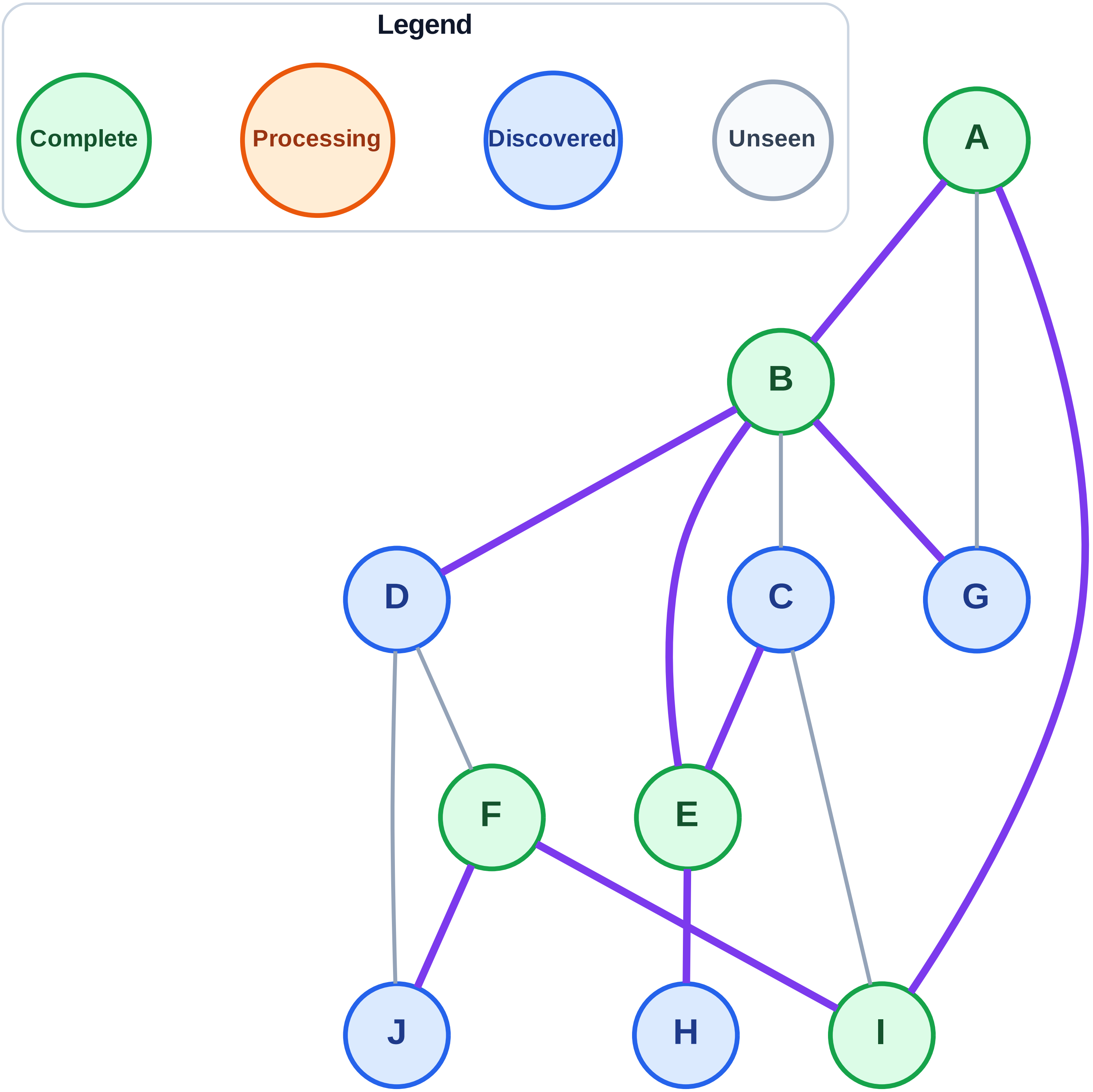
DFS Traversal

Step 10: Process node F



DFS Traversal

Step 11: Discover J from F

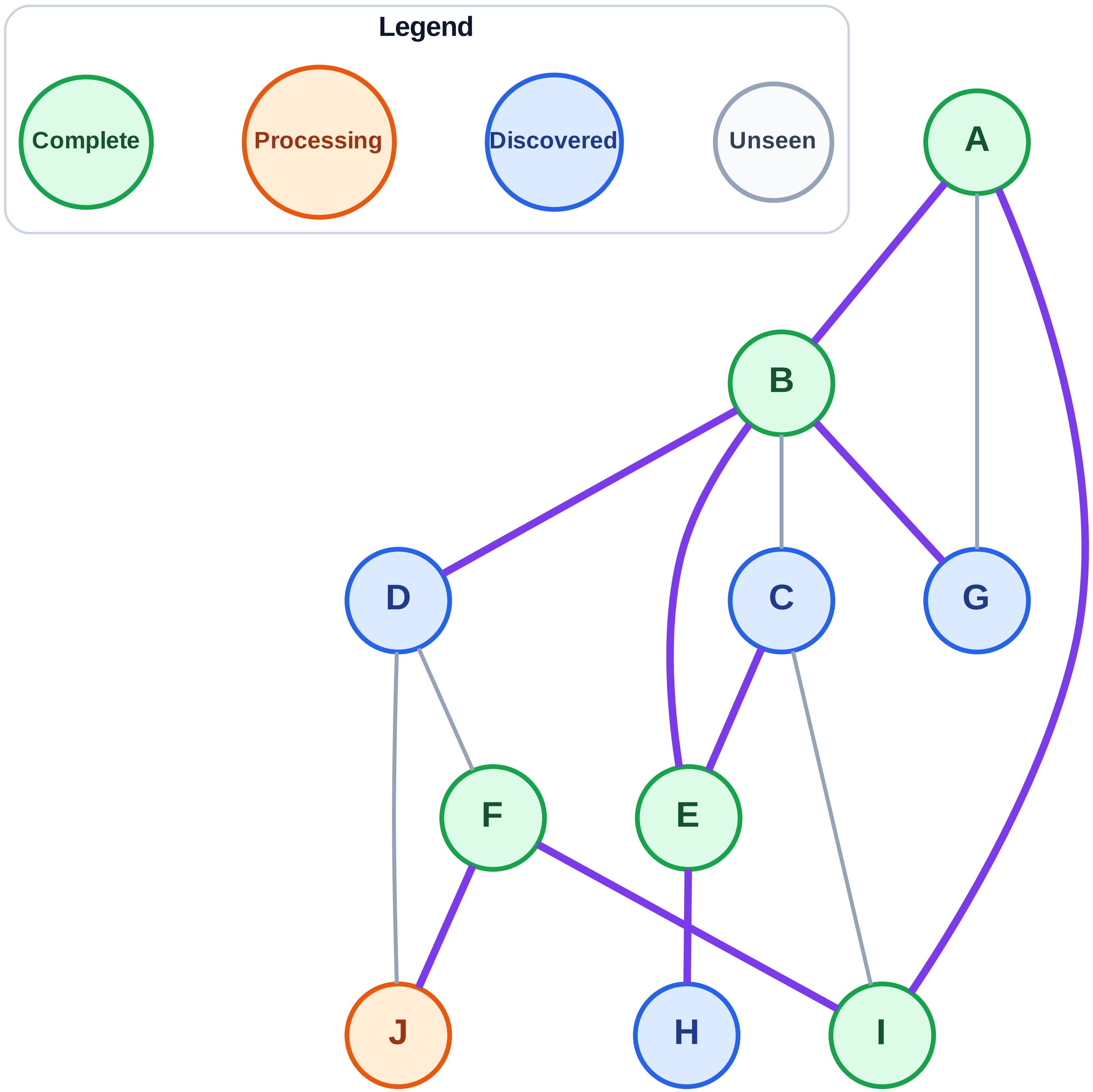


Stack (top at right): H | C | G | D | J

Visited: A, B, E, F, I

DFS Traversal

Step 12: Process node J

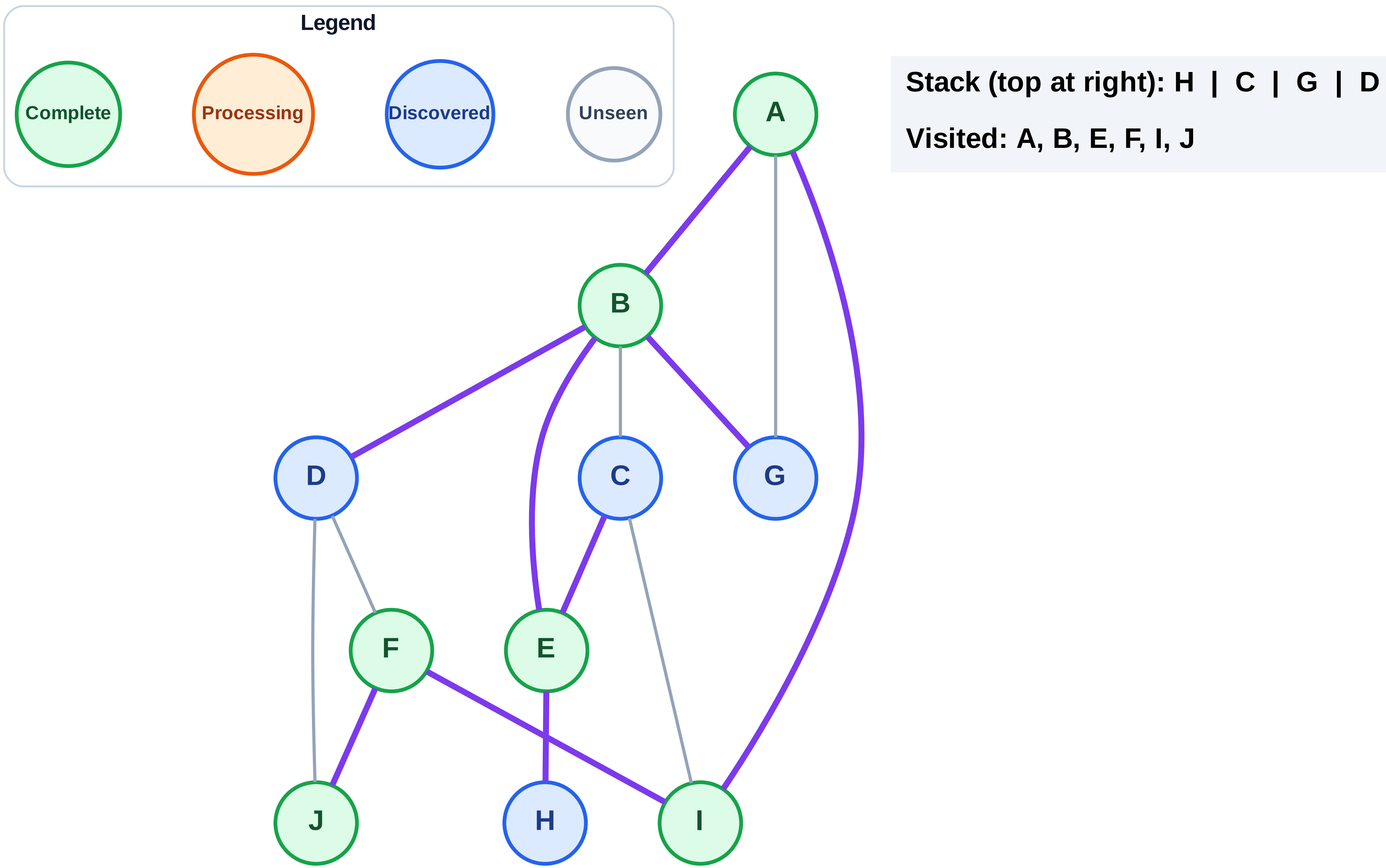


Stack (top at right): H | C | G | D

Visited: A, B, E, F, I

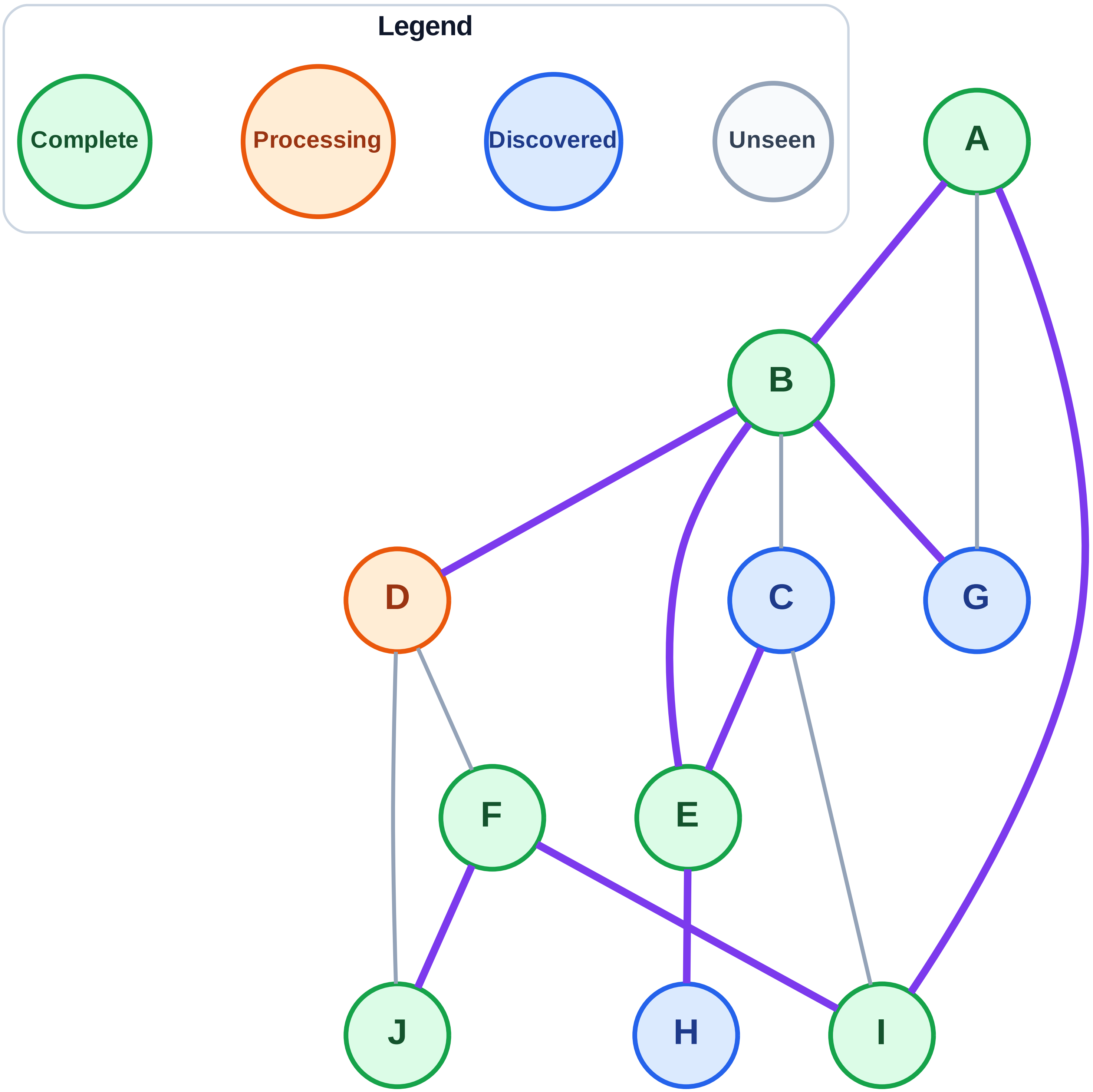
DFS Traversal

Step 13: No new neighbors from J



DFS Traversal

Step 14: Process node D

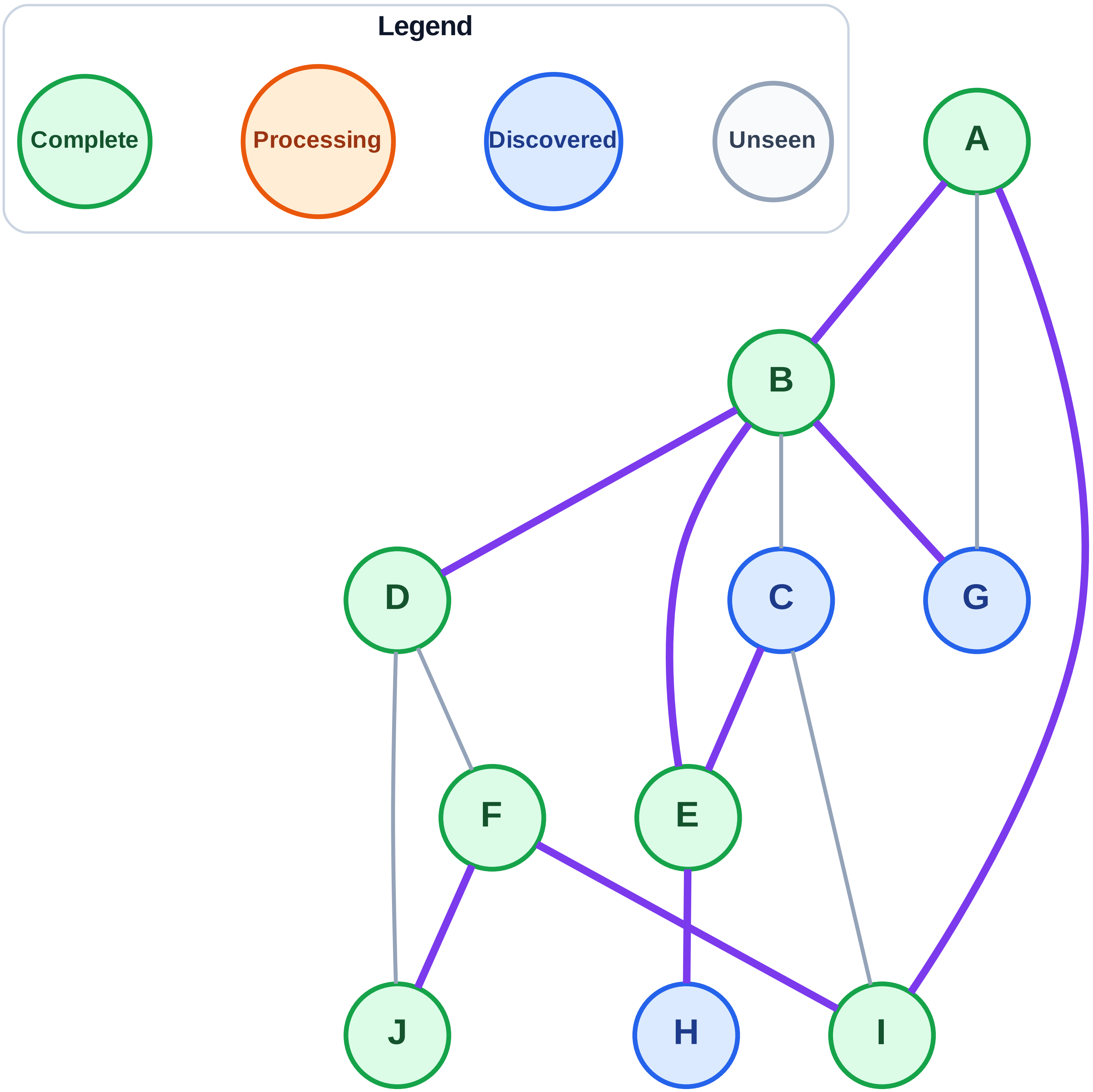


Stack (top at right): H | C | G

Visited: A, B, E, F, I, J

DFS Traversal

Step 15: No new neighbors from D

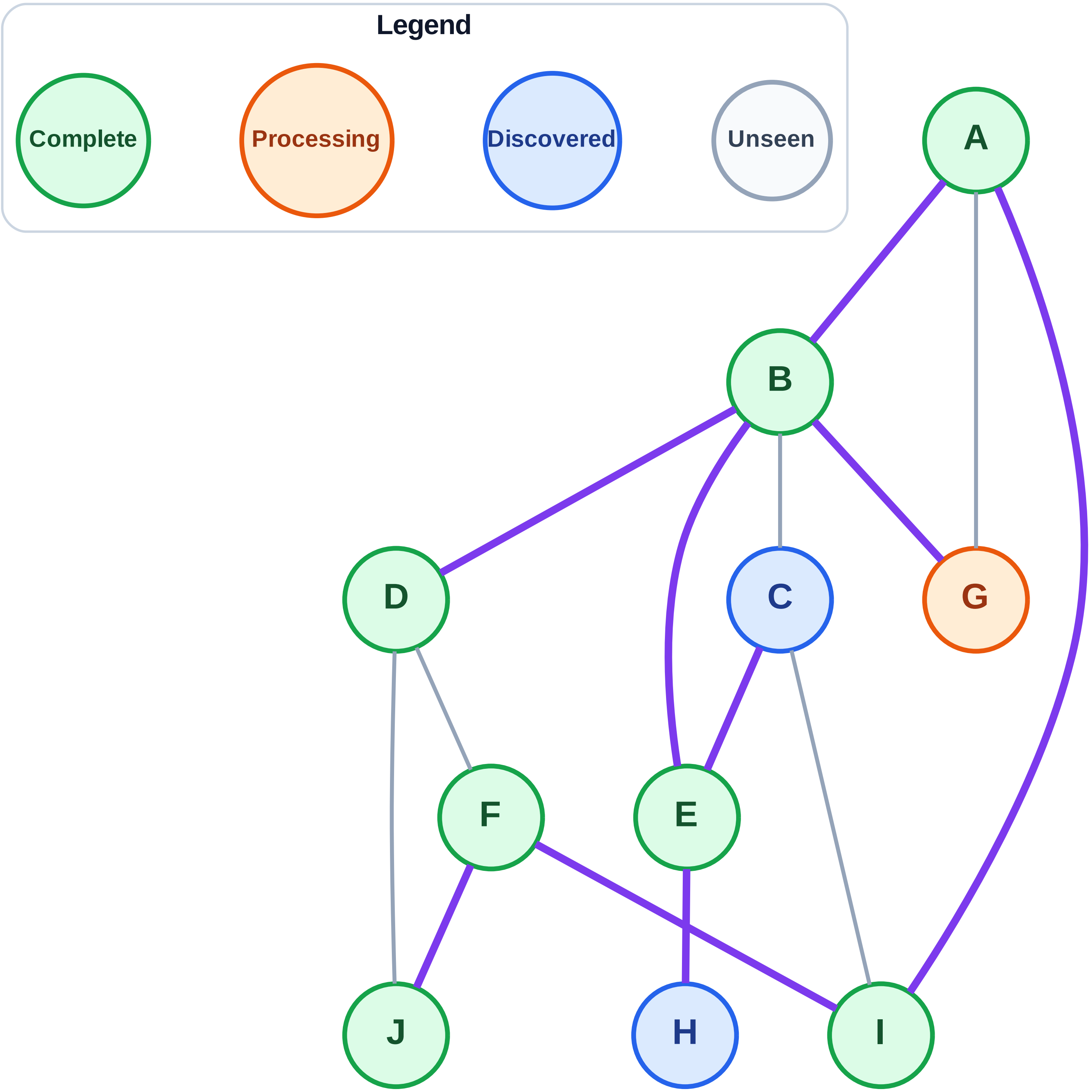


Stack (top at right): H | C | G

Visited: A, B, D, E, F, I, J

DFS Traversal

Step 16: Process node G

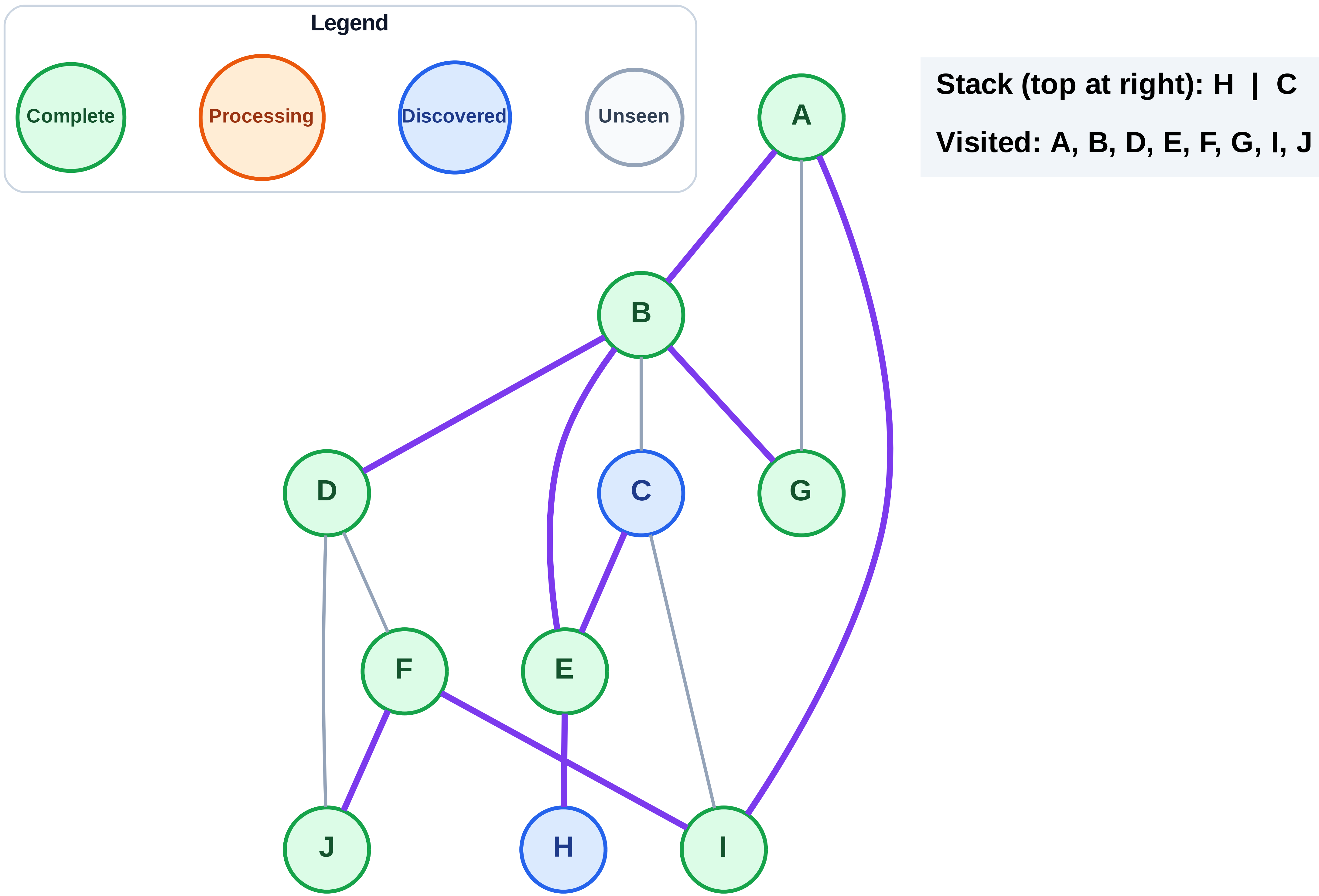


Stack (top at right): H | C

Visited: A, B, D, E, F, I, J

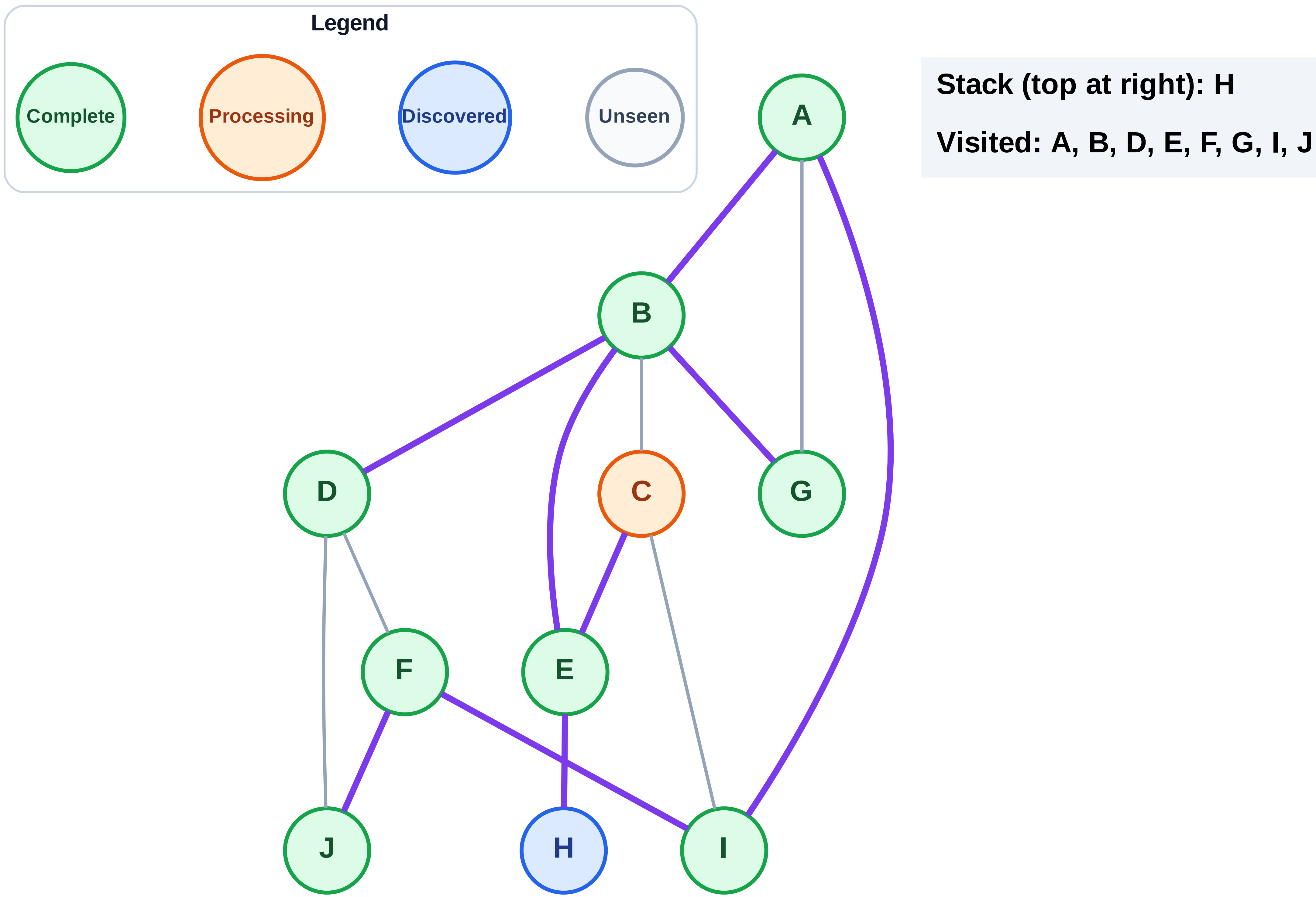
DFS Traversal

Step 17: No new neighbors from G



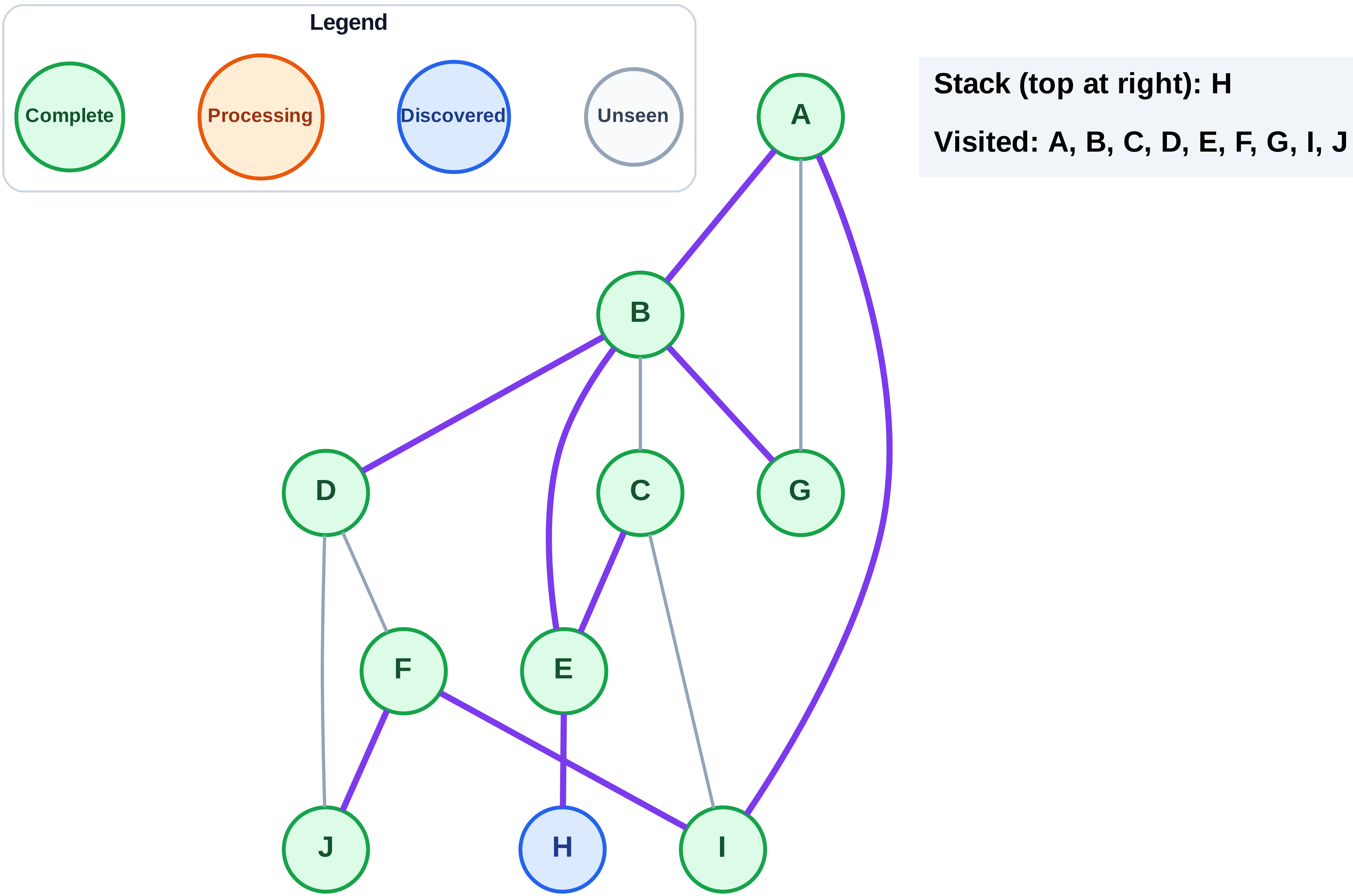
DFS Traversal

Step 18: Process node C



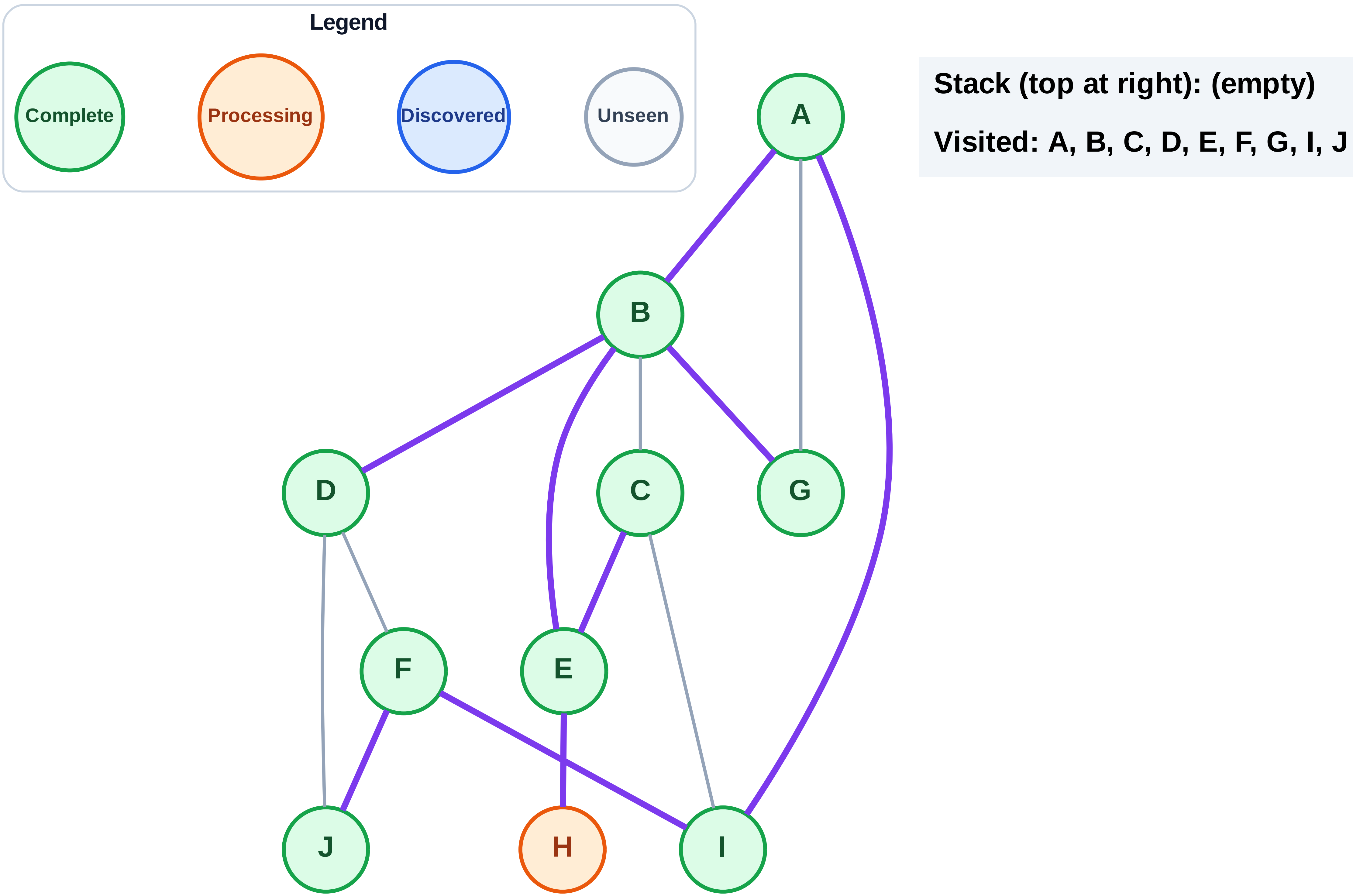
DFS Traversal

Step 19: No new neighbors from C



DFS Traversal

Step 20: Process node H



DFS Traversal

Step 21: No new neighbors from H